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Heterocyclic compound and organic light-emitting device comprising same

Abstract

The present specification relates to a heterocyclic compound represented by Chemical Formula 1, and an organic light emitting device including the same.

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Background/Summary

TECHNICAL FIELD

(1) The present specification relates to a heterocyclic compound, and an organic light emitting device including the same.

(2) This application claims priority to and the benefits of Korean Patent Application No. 10-2019-0122320, filed with the Korean Intellectual Property Office on Oct. 2, 2019, the entire contents of which are incorporated herein by reference.

BACKGROUND ART

(3) An electroluminescent device is one type of self-emissive display devices, and has an advantage of having a wide viewing angle, and a high response speed as well as having an excellent contrast.

(4) An organic light emitting device has a structure disposing an organic thin film between two electrodes. When a voltage is applied to an organic light emitting device having such a structure, electrons and holes injected from the two electrodes bind and pair in the organic thin film, and light emits as these annihilate. The organic thin film may be formed in a single layer or a multilayer as necessary.

(5) A material of the organic thin film may have a light emitting function as necessary. For example, as a material of the organic thin film, compounds capable of forming a light emitting layer themselves alone may be used, or compounds capable of performing a role of a host or a dopant of a host-dopant-based light emitting layer may also be used. In addition thereto, compounds capable of performing roles of hole injection, hole transfer, electron blocking, hole blocking, electron transfer, electron injection and the like may also be used as a material of the organic thin film.

(6) Development of an organic thin film material has been continuously required for enhancing performance, lifetime or efficiency of an organic light emitting device.

DISCLOSURE

Technical Problem

(7) The present specification is directed to providing a heterocyclic compound, and an organic light emitting device including the same.

Technical Solution

(8) One embodiment of the present specification provides a heterocyclic compound represented by the following Chemical Formula 1.

(9) ##STR00001##

(10) In Chemical Formula 1, X is O or S, L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group, Z and R1 are each independently a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2

to C60 heteroarylamino group; or a substituted or unsubstituted phosphine oxide group, R2 to R5 are each independently hydrogen; deuterium; a halogen group; a cyano group; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group, a is an integer of 1 to 4, b is an integer of 1 to 3, m and n are each an integer of 1 to 5, and when a, b, m and n are each 2 or greater, substituents in the parentheses are the same as or different from each other.

(11) Another embodiment of the present application provides an organic light emitting device including a first electrode; a second electrode provided opposite to the first electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound represented by Chemical Formula 1.

(12) Another embodiment of the present application provides an organic light emitting device including a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer includes the heterocyclic compound represented by Chemical Formula 1.

Advantageous Effects

(13) A compound described in the present specification can be used as a material of an organic material layer of an organic light emitting device. In the organic light emitting device, the compound is capable of performing a role of a hole injection material, a hole transfer material, a light emitting material, an electron transfer material, an electron injection material or the like. Particularly, the compound can be used as an electron transfer layer material or a charge generation layer material of an organic light emitting device.

(14) Particularly, Chemical Formula 1 has a structure in which dibenzofuran or dibenzothiophene is fused to quinoline, and by having a substituent in the benzene ring of the quinoline, efficiency is improved by properly controlling HOMO and LUMO distributions and thereby allowing charges to migrate smoothly.

(15) Specifically, when using the compound represented by Chemical Formula 1 in an organic material layer, a driving voltage of the device can be lowered, light efficiency can be enhanced, and lifetime properties of the device can be enhanced.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 to FIG. 5 are diagrams each illustrating a lamination structure of an organic light emitting device according to one embodiment of the present specification. **100**: Substrate **200**: Anode **300**: Organic Material Layer **301**: Hole Injection Layer **302**: Hole Transfer Layer **303**: Light Emitting Layer **304**: Electron Transfer Layer **305**: Electron Injection Layer **306**: Charge Generation Layer **400**: Cathode

MODE FOR DISCLOSURE

(2) Hereinafter, the present specification will be described in more detail.

(3) In the present specification, a certain part “including” certain constituents means capable of further including other constituents, and does not exclude other constituents unless particularly stated on the contrary.

(4) The term “substitution” means a hydrogen atom bonding to a carbon atom of a compound being changed to another substituent, and the position of substitution is not limited as long as it is a position at which the hydrogen atom is substituted, that is, a position at which a substituent can substitute, and when two or more substituents substitute, the two or more substituents may be the

same as or different from each other.

(5) In the present specification, “substituted or unsubstituted” means being substituted with one or more substituents selected from the group consisting of a C1 to C60 linear or branched alkyl group; a C2 to C60 linear or branched alkenyl group; a C2 to C60 linear or branched alkynyl group; a C3 to C60 monocyclic or polycyclic cycloalkyl group; a C2 to C60 monocyclic or polycyclic heterocycloalkyl group; a C6 to C60 monocyclic or polycyclic aryl group; a C2 to C60 monocyclic or polycyclic heteroaryl group; a silyl group; a phosphine oxide group; and an amine group, or being unsubstituted, or being substituted with a substituent linking two or more substituents selected from among the substituents illustrated above, or being unsubstituted.

(6) In the present specification, a “case of a substituent being not indicated in a chemical formula or compound structure” means that a hydrogen atom bonds to a carbon atom. However, since deuterium (²H) is an isotope of hydrogen, some hydrogen atoms may be deuterium.

(7) In one embodiment of the present application, a “case of a substituent being not indicated in a chemical formula or compound structure” may mean that positions that may come as a substituent may all be hydrogen or deuterium. In other words, since deuterium is an isotope of hydrogen, some hydrogen atoms may be deuterium that is an isotope, and herein, a content of the deuterium may be from 0% to 100%.

(8) In one embodiment of the present application, in a “case of a substituent being not indicated in a chemical formula or compound structure”, hydrogen and deuterium may be mixed in compounds when deuterium is not explicitly excluded such as a deuterium content being 0%, a hydrogen content being 100% or substituents being all hydrogen.

(9) In one embodiment of the present application, deuterium is one of isotopes of hydrogen, is an element having deuteron formed with one proton and one neutron as a nucleus, and may be expressed as hydrogen-2, and the elemental symbol may also be written as D or ²H.

(10) In one embodiment of the present application, an isotope means an atom with the same atomic number (Z) but with a different mass number (A), and may also be interpreted as an element with the same number of protons but with a different number of neutrons.

(11) In one embodiment of the present application, a meaning of a content T % of a specific substituent may be defined as $T2/T1 \times 100 = T \%$ when the total number of substituents that a basic compound may have is defined as T1, and the number of specific substituents among these is defined as T2.

(12) In other words, in one example, having a deuterium content of 20% in a phenyl group represented by

(13) ##STR00002##

means that the total number of substituents that the phenyl group may have is 5 (T1 in the formula), and the number of deuterium among these is 1 (T2 in the formula). In other words, having a deuterium content of 20% in a phenyl group may be represented by the following structural formulae.

(14) ##STR00003##

(15) In addition, in one embodiment of the present application, “a phenyl group having a deuterium content of 0%” may mean a phenyl group that does not include a deuterium atom, that is, a phenyl group that has 5 hydrogen atoms.

(16) In the present specification, the halogen may be fluorine, chlorine, bromine or iodine.

(17) In the present specification, the alkyl group includes linear or branched having 1 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkyl group may be from 1 to 60, specifically from 1 to 40 and more specifically from 1 to 20.

Specific examples thereof may include a methyl group, an ethyl group, a propyl group, an n-propyl group, an isopropyl group, a butyl group, an n-butyl group, an isobutyl group, a tert-butyl group, a sec-butyl group, a 1-methyl-butyl group, a 1-ethylbutyl group, a pentyl group, an n-pentyl group, an isopentyl group, a neopentyl group, a tert-pentyl group, a hexyl group, an n-hexyl group, a 1-

methylpentyl group, a 2-methylpentyl group, a 4-methyl-2-pentyl group, a 3,3-dimethylbutyl group, a 2-ethylbutyl group, a heptyl group, an n-heptyl group, a 1-methylhexyl group, a cyclopentylmethyl group, a cyclohexylmethyl group, an octyl group, an n-octyl group, a tert-octyl group, a 1-methylheptyl group, a 2-ethylhexyl group, a 2-propylpentyl group, an n-nonyl group, a 2,2-dimethylheptyl group, a 1-ethyl-propyl group, a 1,1-dimethyl-propyl group, an isohexyl group, a 4-methylhexyl group, a 5-methylhexyl group and the like, but are not limited thereto.

(18) In the present specification, the alkenyl group includes linear or branched having 2 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkenyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 2 to 20. Specific examples thereof may include a vinyl group, a 1-propenyl group, an isopropenyl group, a 1-butenyl group, a 2-butenyl group, a 3-butenyl group, a 1-pentenyl group, a 2-pentenyl group, a 3-pentenyl group, a 3-methyl-1-butenyl group, a 1,3-butadienyl group, an allyl group, a 1-phenylvinyl-1-yl group, a 2-phenylvinyl-1-yl group, a 2,2-diphenylvinyl-1-yl group, a 2-phenyl-2-(naphthyl-1-yl)vinyl-1-yl group, a 2,2-bis(diphenyl-1-yl)vinyl-1-yl group, a stilbenyl group, a styrenyl group and the like, but are not limited thereto.

(19) In the present specification, the alkynyl group includes linear or branched having 2 to 60 carbon atoms, and may be further substituted with other substituents. The number of carbon atoms of the alkynyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 2 to 20.

(20) In the present specification, the alkoxy group may be linear, branched or cyclic. The number of carbon atoms of the alkoxy group is not particularly limited, but is preferably from 1 to 20. Specific examples thereof may include methoxy, ethoxy, n-propoxy, isopropoxy, i-propyloxy, n-butoxy, isobutoxy, tert-butoxy, sec-butoxy, n-pentyloxy, neopentyloxy, isopentyloxy, n-hexyloxy, 3,3-dimethylbutyloxy, 2-ethylbutyloxy, n-octyloxy, n-nonyloxy, n-decyloxy, benzyloxy, p-methylbenzyloxy and the like, but are not limited thereto.

(21) In the present specification, the cycloalkyl group includes monocyclic or polycyclic having 3 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the cycloalkyl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a cycloalkyl group, but may also be different types of cyclic groups such as a heterocycloalkyl group, an aryl group and a heteroaryl group. The number of carbon groups of the cycloalkyl group may be from 3 to 60, specifically from 3 to 40 and more specifically from 5 to 20. Specific examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a 3-methylcyclopentyl group, a 2,3-dimethylcyclopentyl group, a cyclohexyl group, a 3-methylcyclohexyl group, a 4-methylcyclohexyl group, a 2,3-dimethylcyclohexyl group, a 3,4,5-trimethylcyclohexyl group, a 4-tert-butylcyclohexyl group, a cycloheptyl group, a cyclooctyl group and the like, but are not limited thereto.

(22) In the present specification, the heterocycloalkyl group includes O, S, Se, N or Si as a heteroatom, includes monocyclic or polycyclic having 2 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heterocycloalkyl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a heterocycloalkyl group, but may also be different types of cyclic groups such as a cycloalkyl group, an aryl group and a heteroaryl group. The number of carbon atoms of the heterocycloalkyl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 3 to 20.

(23) In the present specification, the aryl group includes monocyclic or polycyclic having 6 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the aryl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be an aryl group, but may also be different types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and a heteroaryl group. The aryl group includes a spiro group. The number of carbon atoms of the aryl group may be from 6 to 60, specifically from 6 to

40 and more specifically from 6 to 25. Specific examples of the aryl group may include a phenyl group, a biphenyl group, a triphenyl group, a naphthyl group, an anthryl group, a chrysenyl group, a phenanthrenyl group, a perylenyl group, a fluoranthenyl group, a triphenylenyl group, a phenalenyl group, a pyrenyl group, a tetracenyl group, a pentacenyl group, a fluorenyl group, an indenyl group, an acenaphthylenyl group, a benzofluorenyl group, a spirobifluorenyl group, a 2,3-dihydro-1H-indenyl group, a fused ring thereof, and the like, but are not limited thereto.

(24) In the present specification, the fluorenyl group may be substituted, and adjacent substituents may bond to each other to form a ring.

(25) When the fluorenyl group is substituted,

(26) ##STR00004##

and the like may be included, however, the structure is not limited thereto.

(27) In the present specification, the heteroaryl group includes S, O, Se, N or Si as a heteroatom, includes monocyclic or polycyclic having 2 to 60 carbon atoms, and may be further substituted with other substituents. Herein, the polycyclic means a group in which the heteroaryl group is directly linked to or fused with other cyclic groups. Herein, the other cyclic groups may be a heteroaryl group, but may also be different types of cyclic groups such as a cycloalkyl group, a heterocycloalkyl group and an aryl group. The number of carbon atoms of the heteroaryl group may be from 2 to 60, specifically from 2 to 40 and more specifically from 3 to 25. Specific examples of the heteroaryl group may include a pyridyl group, a pyrrolyl group, a pyrimidyl group, a pyridazinyl group, a furanyl group, a thiophene group, an imidazolyl group, a pyrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, a triazolyl group, a furazanyl group, an oxadiazolyl group, a thiadiazolyl group, a dithiazolyl group, a tetrazolyl group, a pyranlyl group, a thiopyranlyl group, a diazinyl group, an oxazinyl group, a thiazinyl group, a dioxynyl group, a triazinyl group, a tetrazinyl group, a quinolyl group, an isoquinolyl group, a quinazolinyl group, an isoquinazolinyl group, a quinoxalinyl group, a naphthyridyl group, an acridinyl group, a phenanthridinyl group, an imidazopyridinyl group, a diazanaphthalenyl group, a triazaindene group, an indolyl group, an indolizinyl group, a benzothiazolyl group, a benzoxazolyl group, a benzimidazolyl group, a benzothiophene group, a benzofuran group, a dibenzothiophene group, a dibenzofuran group, a carbazolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a phenazinyl group, a dibenzosilole group, spirobi(dibenzosilole), a dihydrophenazinyl group, a phenoxazinyl group, a phenanthridyl group, an imidazopyridinyl group, a thienyl group, an indolo[2,3-a]carbazolyl group, an indolo[2,3-b]carbazolyl group, an indolynyl group, a 10,11-dihydro-dibenzo[b,f]azepine group, a 9,10-dihydroacridinyl group, a phenanthrazinyl group, a phenothiathiazinyl group, a phthalazinyl group, a naphthylidinyl group, a phenanthrolinyl group, a benzo[c][1,2,5]thiadiazolyl group, a 5,10-dihydrobenzo[b,e][1,4]azasilinyl group, a pyrazolo[1,5-c]quinazolinyl group, a pyrido[1,2-b]indazolyl group, a pyrido[1,2-a]imidazo[1,2-e]indolynyl group, a 5,11-dihydroindeno[1,2-b]carbazolyl group and the like, but are not limited thereto.

(28) In the present specification, the silyl group is a substituent including Si, having the Si atom directly linked as a radical, and is represented by —Si(R101) (R102) (R103). R101 to R103 are the same as or different from each other, and may be each independently a substituent formed with at least one of hydrogen; deuterium; a halogen group; an alkyl group; an alkenyl group; an alkoxy group; a cycloalkyl group; a heterocycloalkyl group; an aryl group; and a heteroaryl group.

Specific examples of the silyl group may include a trimethylsilyl group, a triethylsilyl group, a t-butyltrimethylsilyl group, a vinyltrimethylsilyl group, a propyltrimethylsilyl group, a triphenylsilyl group, a diphenylsilyl group, a phenylsilyl group and the like, but are not limited thereto.

(29) In the present specification, the phosphine oxide group is represented by —P(=O) (R104)

(R105), and R104 and R105 are the same as or different from each other and may be each

independently a substituent formed with at least one of hydrogen; deuterium; a halogen group; an alkyl group; an alkenyl group; an alkoxy group; a cycloalkyl group; a heterocycloalkyl group; an aryl group; and a heteroaryl group. The phosphine oxide group may specifically be substituted with

an aryl group, and as the aryl group, the examples described above may be applied. Examples of the phosphine oxide group may include a diphenylphosphine oxide group, a dinaphthylphosphine oxide group and the like, but are not limited thereto.

(30) In the present specification, the amine group is represented by —N(R106) (R107), and R106 and R107 are the same as or different from each other and may be each independently a substituent formed with at least one of hydrogen; deuterium; a halogen group; an alkyl group; an alkenyl group; an alkoxy group; a cycloalkyl group; a heterocycloalkyl group; an aryl group; and a heteroaryl group. The amine group may be selected from the group consisting of —NH₂; a monoalkylamine group; a monoarylamine group; a monoheteroarylamine group; a dialkylamine group; a diarylamine group; a diheteroarylamine group; an alkylarylamine group; an alkylheteroarylamine group; and an arylheteroarylamine group, and although not particularly limited thereto, the number of carbon atoms is preferably from 1 to 30. Specific examples of the amine group may include a methylamine group, a dimethylamine group, an ethylamine group, a diethylamine group, a phenylamine group, a naphthylamine group, a biphenylamine group, a dibiphenylamine group, an anthracenylamine group, a 9-methyl-anthracenylamine group, a diphenylamine group, a phenylnaphthylamine group, a ditolylamine group, a phenyltolylamine group, a triphenylamine group, a biphenylnaphthylamine group, a phenylbiphenylamine group, a biphenylfluorenylamine group, a phenyltriphenylenylamine group, a biphenyltriphenylenylamine group and the like, but are not limited thereto.

(31) In the present specification, descriptions on the aryl group provided above may be applied to the arylene group except that the arylene group is not a monovalent group.

(32) In the present specification, descriptions on the heteroaryl group provided above may be applied to the heteroarylene group except that the heteroarylene group is not a monovalent group.

(33) In the present specification, an “adjacent” group may mean a substituent substituting an atom directly linked to an atom substituted by the corresponding substituent, a substituent sterically most closely positioned to the corresponding substituent, or another substituent substituting an atom substituted by the corresponding substituent. For example, two substituents substituting ortho positions in a benzene ring, and two substituents substituting the same carbon in an aliphatic ring may be interpreted as groups “adjacent” to each other.

(34) As the aliphatic or aromatic hydrocarbon ring or heteroring that the adjacent groups may form, the structures illustrated as the cycloalkyl group, the heterocycloalkyl group, the aryl group and the heteroaryl group described above may be applied except for those that are not a monovalent group.

(35) One embodiment of the present specification provides a heterocyclic compound represented by Chemical Formula 1.

(36) Particularly, Chemical Formula 1 has a structure in which dibenzofuran or dibenzothiophene is fused to quinoline, and by having a substituent in the benzene ring of the quinoline, HOMO and LUMO tend to have a decreased overlap, and light emission efficiency is improved. In addition, through properly controlling an energy level, a migration barrier of charges is lowered, which improves device efficiency. In addition, by attaching a substituent having HT properties, particularly an aryl group, to R1 of the ring including N of the quinoline, the shape of the molecule may be improved, and thermal stability of the molecule may increase therethrough.

(37) In one embodiment of the present specification, X is O or S.

(38) In one embodiment of the present specification, X is O.

(39) In another embodiment, X is S.

(40) In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted C6 to C60 arylene group; or a substituted or unsubstituted C2 to C60 heteroarylene group.

(41) In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted C6 to C30 arylene group; or a substituted or unsubstituted C2 to C30 heteroarylene group.

(42) In one embodiment of the present specification, L is a direct bond; or a substituted or unsubstituted C6 to C30 arylene group.

(43) In one embodiment of the present specification, L is a direct bond; a substituted or unsubstituted phenylene group; a substituted or unsubstituted biphenylene group; a substituted or unsubstituted terphenylene group; a substituted or unsubstituted naphthylene group; a substituted or unsubstituted anthracenylene group; a substituted or unsubstituted triphenylenylene group; or a substituted or unsubstituted phenanthrenylene group.

(44) In one embodiment of the present specification, L is a direct bond; a phenylene group; a biphenylene group; a terphenylene group; a naphthylene group; an anthracenylene group; a triphenylenylene group; or a phenanthrenylene group.

(45) In one embodiment of the present specification, L is a direct bond; or a phenylene group.

(46) In one embodiment of the present specification, Z is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

(47) In one embodiment of the present specification, Z is a substituted or unsubstituted C1 to C30 alkyl group; a substituted or unsubstituted C2 to C30 alkenyl group; a substituted or unsubstituted C2 to C30 alkynyl group; a substituted or unsubstituted C3 to C30 cycloalkyl group; a substituted or unsubstituted C2 to C30 heterocycloalkyl group; a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; a substituted or unsubstituted C1 to C10 alkylamine group; a substituted or unsubstituted C6 to C30 arylamine group; a substituted or unsubstituted C2 to C30 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

(48) In one embodiment of the present specification, Z is a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group; or a substituted or unsubstituted phosphine oxide group.

(49) In one embodiment of the present specification, Z is a substituted or unsubstituted phenyl group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a benzonaphthothiophene group; a substituted or unsubstituted pyrazole group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted triazine group; a substituted or unsubstituted carbazole group; a substituted or unsubstituted phenanthrolinyl group; a substituted or unsubstituted dibenzofuran group; a substituted or unsubstituted dibenzothiophene group; a substituted or unsubstituted dihydroacridine group; or a substituted or unsubstituted phosphine oxide group.

(50) In one embodiment of the present specification, Z is a phenyl group unsubstituted or substituted with a cyano group, a phenyl group or a carbazole group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a benzonaphthothiophene group; a pyrazole group unsubstituted or substituted with a phenyl group; a pyridine group unsubstituted or substituted with a phenyl group or a pyridine group; a pyrimidine group unsubstituted or substituted with a phenyl group or a biphenyl group; a triazine group unsubstituted or substituted with one or more substituents selected from the group consisting

of a phenyl group unsubstituted or substituted with a naphthyl group or a cyano group, a biphenyl group, a naphthyl group, a dimethylfluorenyl group, a dibenzofuran group, a dibenzothiophene group and a carbazole group unsubstituted or substituted with a phenyl group; a carbazole group unsubstituted or substituted with a phenyl group; a phenanthrolinyl group unsubstituted or substituted with a phenyl group; a dibenzofuran group unsubstituted or substituted with a phenyl group or a naphthyl group; a dibenzothiophene group unsubstituted or substituted with a phenyl group or a naphthyl group; a dihydroacridine group unsubstituted or substituted with a methyl group or a phenyl group; or a phosphine oxide group unsubstituted or substituted with a phenyl group.

(51) In one embodiment of the present specification, R1 is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C2 to C60 alkenyl group; a substituted or unsubstituted C2 to C60 alkynyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; a substituted or unsubstituted C2 to C60 heteroaryl group; a substituted or unsubstituted C1 to C20 alkylamine group; a substituted or unsubstituted C6 to C60 arylamine group; a substituted or unsubstituted C2 to C60 heteroarylamine group; or a substituted or unsubstituted phosphine oxide group.

(52) In one embodiment of the present specification, R1 is a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C2 to C60 heterocycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(53) In one embodiment of the present specification, R1 is a C6 to C60 substituted or unsubstituted aryl group; or a C2 to C60 substituted or unsubstituted heteroaryl group.

(54) In one embodiment of the present specification, R1 is a C6 to C30 substituted or unsubstituted aryl group; or a C2 to C30 substituted or unsubstituted heteroaryl group.

(55) In one embodiment of the present specification, R1 is a C6 to C30 substituted or unsubstituted aryl group.

(56) In one embodiment of the present specification, R1 is a substituted or unsubstituted phenyl group; a substituted or unsubstituted naphthyl group; a substituted or unsubstituted anthracenyl group; a substituted or unsubstituted phenanthrenyl group; a substituted or unsubstituted pyrenyl group; or a substituted or unsubstituted perylene group.

(57) In one embodiment of the present specification, R1 is a phenyl group unsubstituted or substituted with a naphthyl group; a naphthyl group; an anthracenyl group; a phenanthrenyl group; a pyrenyl group; or a perylene group.

(58) In one embodiment of the present specification, R2 to R5 are each independently hydrogen; deuterium; a halogen group; a cyano group; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C3 to C60 cycloalkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(59) In one embodiment of the present specification, R2 to R5 are each independently hydrogen; deuterium; a substituted or unsubstituted C1 to C60 alkyl group; a substituted or unsubstituted C6 to C60 aryl group; or a substituted or unsubstituted C2 to C60 heteroaryl group.

(60) In one embodiment of the present specification, R2 to R5 are each independently hydrogen; or deuterium.

(61) In one embodiment of the present specification, R2 to R5 are hydrogen.

(62) In one embodiment of the present specification, m may be 1 or 2.

(63) In one embodiment of the present specification, n may be an integer of 1 to 3.

(64) In the heterocyclic compound provided in one embodiment of the present specification, Chemical Formula 1 is represented by the following Chemical Formula 2 or 3.

(65) ##STR00005##

(66) In Chemical Formulae 2 and 3, L, Z, R1, R4, R5, a, b, m and n have the same definitions as in

Chemical Formula 1.

(67) In the heterocyclic compound provided in one embodiment of the present specification, Chemical Formula 1 is represented by any one of the following Chemical Formulae 4 to 6.

(68) ##STR00006##

(69) In Chemical Formulae 4 to 6, X, L, Z, R1 to R5, a, b, m and n have the same definitions as in Chemical Formula 1.

(70) In one embodiment of the present specification, Chemical Formula 1 may be represented by any one of the following compounds, but is not limited thereto.

(71) ##STR00007## ##STR00008## ##STR00009## ##STR00010## ##STR00011##
##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017##
##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##
##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029##
##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035##
##STR00036##

(72) ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##
##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047##
##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053##
##STR00054## ##STR00055## ##STR00056## ##STR00057## ##STR00058## ##STR00059##
##STR00060## ##STR00061## ##STR00062## ##STR00063## ##STR00064## ##STR00065##
##STR00066## ##STR00067## ##STR00068## ##STR00069## ##STR00070## ##STR00071##
##STR00072## ##STR00073## ##STR00074## ##STR00075## ##STR00076## ##STR00077##
##STR00078## ##STR00079## ##STR00080## ##STR00081## ##STR00082## ##STR00083##
##STR00084## ##STR00085## ##STR00086## ##STR00087##

(73) ##STR00088## ##STR00089## ##STR00090## ##STR00091## ##STR00092##
##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098##
##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104##
##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110##
##STR00111## ##STR00112## ##STR00113##

(74) ##STR00114## ##STR00115## ##STR00116## ##STR00117## ##STR00118##
##STR00119## ##STR00120## ##STR00121## ##STR00122## ##STR00123## ##STR00124##
##STR00125## ##STR00126## ##STR00127## ##STR00128## ##STR00129## ##STR00130##
##STR00131## ##STR00132## ##STR00133## ##STR00134## ##STR00135## ##STR00136##
##STR00137## ##STR00138## ##STR00139## ##STR00140## ##STR00141## ##STR00142##
##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148##

(75) ##STR00149## ##STR00150## ##STR00151## ##STR00152## ##STR00153##
##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158## ##STR00159##
##STR00160## ##STR00161## ##STR00162## ##STR00163## ##STR00164## ##STR00165##
##STR00166## ##STR00167## ##STR00168## ##STR00169## ##STR00170## ##STR00171##
##STR00172## ##STR00173## ##STR00174## ##STR00175## ##STR00176## ##STR00177##
##STR00178## ##STR00179## ##STR00180## ##STR00181## ##STR00182## ##STR00183##
##STR00184## ##STR00185## ##STR00186## ##STR00187## ##STR00188## ##STR00189##
##STR00190## ##STR00191## ##STR00192##

(76) In addition, by introducing various substituents to the structure of Chemical Formula 1, compounds having unique properties of the introduced substituents may be synthesized. For example, by introducing substituents normally used as hole injection layer materials, hole transfer layer materials, light emitting layer materials, electron transfer layer materials and charge generation layer materials used for manufacturing an organic light emitting device to the core structure, materials satisfying conditions required for each organic material layer may be synthesized. In addition, the energy band gap may be finely controlled, and with a high T1 value obtained therefrom, an organic light emitting device with superior efficiency may be provided.

Herein, the T1 value means an energy level value in a triplet state.

(77) In addition, by introducing various substituents with different properties to the structure of Chemical Formula 1, efficiency may increase through controlling HOMO and LUMO, and interfacial properties may be improved through controlling the molecular weight.

(78) One embodiment of the present specification provides an organic light emitting device including a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound represented by Chemical Formula 1.

(79) In one embodiment of the present specification, the first electrode may be an anode, and the second electrode may be a cathode.

(80) In another embodiment of the present specification, the first electrode may be a cathode, and the second electrode may be an anode.

(81) In one embodiment of the present specification, the organic light emitting device may be a blue organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the blue organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a charge generation layer or an electron transfer layer of the blue organic light emitting device.

(82) In another embodiment of the present specification, the organic light emitting device may be a green organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the green organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a charge generation layer or an electron transfer layer of the green organic light emitting device.

(83) In another embodiment of the present specification, the organic light emitting device may be a red organic light emitting device, and the heterocyclic compound according to Chemical Formula 1 may be used as a material of the red organic light emitting device. For example, the heterocyclic compound according to Chemical Formula 1 may be included in a charge generation layer or an electron transfer layer of the red organic light emitting device.

(84) Specific descriptions on the heterocyclic compound represented by Chemical Formula 1 are the same as the descriptions provided above.

(85) The organic light emitting device of the present specification may be manufactured using common organic light emitting device manufacturing methods and materials except that one or more of the organic material layers are formed using the heterocyclic compound described above.

(86) The heterocyclic compound may be formed into an organic material layer through a solution coating method as well as a vacuum deposition method when manufacturing the organic light emitting device. Herein, the solution coating method means spin coating, dip coating, inkjet printing, screen printing, a spray method, roll coating and the like, but is not limited thereto.

(87) The organic material layer of the organic light emitting device of the present specification may be formed in a single layer structure, but may be formed in a multilayer structure in which two or more organic material layers are laminated. For example, the organic light emitting device of the present disclosure may have a structure including a hole injection layer, a hole transfer layer, a light emitting layer, an electron transfer layer, an electron injection layer and the like as the organic material layer. However, the structure of the organic light emitting device is not limited thereto, and may include a smaller number of organic material layers.

(88) In the organic light emitting device of the present specification, the organic material layer includes a charge generation layer, and the charge generation layer may include the heterocyclic compound of Chemical Formula 1. When including the heterocyclic compound in the charge generation layer, the band gap may be readily controlled facilitating carrier migration, and thermal properties may be improved by increasing the molecular weight.

(89) In the organic light emitting device of the present specification, the organic material layer includes an electron transfer layer, and the electron transfer layer may include the heterocyclic

compound of Chemical Formula 1. When using the heterocyclic compound as an electron transfer material, HOMO and LUMO may be controlled depending on the substituent position, and electron transfer efficiency is superior. In addition, a higher T1 is obtained compared to existing electron transfer materials leading to a low light emitter loss, and by preventing electrons from migrating in a reverse direction, excellent device efficiency is obtained.

(90) The organic light emitting device of the present disclosure may further include one, two or more layers selected from the group consisting of a light emitting layer, a hole injection layer, a hole transfer layer, an electron injection layer, an electron transfer layer, an electron blocking layer and a hole blocking layer.

(91) FIG. 1 to FIG. 5 illustrate a lamination order of electrodes and organic material layers of an organic light emitting device according to one embodiment of the present specification. However, the scope of the present application is not limited to these diagrams, and structures of organic light emitting devices known in the art may also be used in the present application.

(92) FIG. 1 illustrates an organic light emitting device in which an anode (200), an organic material layer (300) and a cathode (400) are consecutively laminated on a substrate (100). However, the structure is not limited to such a structure, and as illustrated in FIG. 2, an organic light emitting device in which a cathode, an organic material layer and an anode are consecutively laminated on a substrate may also be obtained.

(93) FIG. 3 and FIG. 4 illustrate cases of the organic material layer being a multilayer. The organic light emitting device according to FIG. 3 includes a hole injection layer (301), a hole transfer layer (302), a light emitting layer (303), an electron transfer layer (304) and a charge generation layer (306), and the organic light emitting device according to FIG. 4 includes a hole injection layer (301), a hole transfer layer (302), a light emitting layer (303), an electron transfer layer (304) and an electron injection layer (305). However, the scope of the present application is not limited to such a lamination structure, and as necessary, layers other than the light emitting layer may not be included, and other necessary functional layers may be further added.

(94) The organic material layer including the heterocyclic compound represented by Chemical Formula 1 may further include other materials as necessary.

(95) The organic light emitting device according to one embodiment of the present specification includes a first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a second light emitting layer; and a second electrode provided on the second stack, wherein the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(96) The organic light emitting device according to one embodiment of the present specification includes a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes two or more stacks, and the two or more stacks each independently include a light emitting layer, a charge generation layer is included between the two or more stacks, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(97) The organic light emitting device according to one embodiment of the present specification includes a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes a first stack including a first light emitting layer; a charge generation layer provided on the first stack; and a second stack including a second light emitting layer, and the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1.

(98) In addition, the organic light emitting device according to one embodiment of the present specification includes an anode, a cathode, and two or more stacks provided between the anode and the cathode, the two or more stacks each independently include a light emitting layer, a charge generation layer is included between the two or more stacks, and the charge generation layer

includes the heterocyclic compound represented by Chemical Formula 1.

(99) In addition, the organic light emitting device according to one embodiment of the present specification includes an anode, a first stack provided on the anode and including a first light emitting layer, a charge generation layer provided on the first stack, a second stack provided on the charge generation layer and including a second light emitting layer, and a cathode provided on the second stack. Herein, the charge generation layer may include the heterocyclic compound represented by Chemical Formula 1. When the heterocyclic compound is included in the charge generation layer and used, superior efficiency is obtained as a charge generation layer material due to an electron-friendly substituent structure and a fused structure of quinoline-dibenzofuran or quinoline-dibenzothiophene friendly for hole migration.

(100) Specifically, in the charge generation layer, controlling an energy level is important for an electron transfer layer and a hole transfer layer to favorably receive electrons and holes, respectively. By properly mixing a moiety with ET properties and a moiety with HT properties, the compound of the present disclosure is capable of controlling an energy level through properly controlling a substituent to have a deep LUMO level or a shallow HOMO level, and high efficiency as a charge generation layer may be obtained therethrough.

(101) In addition, the first stack and the second stack may each independently further include one or more types of the hole injection layer, the hole transfer layer, the hole blocking layer, the electron transfer layer, the electron injection layer and the like described above.

(102) The charge generation layer includes an N-type charge generation layer, and the N-type charge generation layer includes the heterocyclic compound represented by Chemical Formula 1 and, in addition to the compound, may further include a dopant known in the art.

(103) As the organic light emitting device according to one embodiment of the present specification, an organic light emitting device having a 2-stack tandem structure is illustrated in FIG. 5.

(104) Herein, the first electron blocking layer, the first hole blocking layer, the second hole blocking layer and the like described in FIG. 5 may not be included in some cases.

(105) In the organic light emitting device according to one embodiment of the present specification, materials other than the heterocyclic compound represented by Chemical Formula 1 are illustrated below, however, these are for illustrative purposes only and not for limiting the scope of the present application, and may be replaced by materials known in the art.

(106) As the anode material, materials having relatively large work function may be used, and transparent conductive oxides, metals, conductive polymers or the like may be used. Specific examples of the anode material include metals such as vanadium, chromium, copper, zinc and gold, or alloys thereof; metal oxides such as zinc oxide, indium oxide, indium tin oxide (ITO) and indium zinc oxide (IZO); combinations of metals and oxides such as ZnO:Al or SnO₂:Sb; conductive polymers such as poly(3-methylthiophene), poly[3,4-(ethylene-1,2-dioxy)thiophene] (PEDOT), polypyrrole and polyaniline, and the like, but are not limited thereto.

(107) As the cathode material, materials having relatively small work function may be used, and metals, metal oxides, conductive polymers or the like may be used. Specific examples of the cathode material include metals such as magnesium, calcium, sodium, potassium, titanium, indium, yttrium, lithium, gadolinium, aluminum, silver, tin and lead, or alloys thereof; multilayer structure materials such as LiF/Al or LiO₂/Al, and the like, but are not limited thereto.

(108) As the hole injection material, known hole injection materials may be used, and for example, phthalocyanine compounds such as copper phthalocyanine disclosed in U.S. Pat. No. 4,356,429, or starburst-type amine derivatives such as tris(4-carbazoyl-9-ylphenyl)amine (TCTA), 4,4',4''-tri[phenyl(m-tolyl)amino]triphenylamine (m-MTDATA) or 1,3,5-tris[4-(3-methylphenylphenylamino)phenyl]benzene (m-MTDAPB) described in the literature [Advanced Material, 6, p. 677 (1994)], polyaniline/dodecylbenzene sulfonic acid, poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate), polyaniline/camphor sulfonic acid or

polyaniline/poly(4-styrenesulfonate) that are conductive polymers having solubility, and the like, may be used.

(109) As the hole transfer material, pyrazoline derivatives, arylamine-based derivatives, stilbene derivatives, triphenyldiamine derivatives and the like may be used, and low molecular or high molecular materials may also be used.

(110) As the electron transfer material, metal complexes of oxadiazole derivatives, anthraquinodimethane and derivatives thereof, benzoquinone and derivatives thereof, naphthoquinone and derivatives thereof, anthraquinone and derivatives thereof, tetracyanoanthraquinodimethane and derivatives thereof, fluorenone derivatives, diphenyldicyanoethylene and derivatives thereof, diphenoquinone derivatives, 8-hydroxyquinoline and derivatives thereof, and the like, may be used in addition to the heterocyclic compound, and high molecular materials may also be used as well as low molecular materials.

(111) As examples of the electron injection material, LiF is typically used in the art, however, the present application is not limited thereto.

(112) As the light emitting material, red, green or blue light emitting materials may be used, and as necessary, two or more light emitting materials may be mixed and used. Herein, two or more light emitting materials may be used by being deposited as individual sources of supply or by being premixed and deposited as one source of supply. In addition, fluorescent materials may also be used as the light emitting material, however, phosphorescent materials may also be used. As the light emitting material, materials emitting light by bonding electrons and holes injected from an anode and a cathode, respectively, may be used alone, however, materials having a host material and a dopant material involving in light emission together may also be used.

(113) When mixing light emitting material hosts, same series hosts may be mixed, or different series hosts may be mixed. For example, any two or more types of materials among n-type host materials or p-type host materials may be selected and used as a host material of a light emitting layer.

(114) The organic light emitting device according to one embodiment of the present specification may be a top-emission type, a bottom-emission type or a dual-emission type depending on the materials used.

(115) The heterocyclic compound according to one embodiment of the present specification may also be used in an organic electronic device including an organic solar cell, an organic photo conductor, an organic transistor and the like under a similar principle used in the organic light emitting device.

(116) Hereinafter, the present specification will be described in more detail with reference to examples, however, these are for illustrative purposes only, and the scope of the present application is not limited thereto.

<Preparation Example 1> Preparation of Compound 1

(117) ##STR00193## ##STR00194##

1) Preparation of Compound 1-3

(118) After dissolving dibenzo[b,d]furan-1-ylboronic acid (30 g, 0.14 mol, 1 eq.) and 2,6-dibromoaniline (52.5 g, 0.21 mol, 1.5 eq.) in 1,4-dioxane (300 ml) and H₂O (90 ml), Pd(PPh₃)₄ (8 g, 0.007 mol, 0.05 eq.) and K₃PO₄ (89 g, 0.42 mol, 3 eq.) were introduced thereto, and the result was stirred for 4 hours under reflux.

(119) The reaction solution was concentrated, then dissolved by introducing methylene chloride (MC) thereto, and then extracted with water, and the organic layer was dried with anhydrous Na₂SO₄. The solution was concentrated and dissolved again in methylene chloride, and after adsorbing the result to silica gel (250 ml), an MC/hexane column was performed to obtain Compound 1-3 (36 g).

2) Preparation of Compound 1-2

(120) After dissolving Compound 1-3 (36 g, 0.11 mol, 1 eq.) and benzaldehyde (17.5 g, 0.17 mol,

1.5 eq.) in 1,2-dichlorobenzene (360 ml) and trifluoromethanesulfonic acid (20 ml), the result was stirred for 3 hours under reflux.

(121) NaHCO₃ dissolved in H₂O was introduced to the reaction solution for neutralization. The solution was concentrated and dissolved in a small amount of MC, and then MC/methanol (MeOH) slurried and filtered. Compound 1-2 (28 g) was obtained in a 60% yield.

3) Preparation of Compound 1-1

(122) After dissolving Compound 1-2 (28 g, 0.066 mol, 1 eq.) and bis(pinacolato)diboron (25 g, 0.099 mol, 1.5 eq.) in 1,4-dioxane (300 ml), Pd(dppf)Cl₂ (2.4 g, 0.033 mol, 0.05 eq.) and KOAc (19 g, 0.198 mol, 3 eq.) were introduced thereto, and the result was stirred for 4 hours under reflux. After the reaction was completed, the result was extracted with MC and water, and then the organic layer was dried with anhydrous MgSO₄ and then silica gel filtered. The result was precipitated with MC/MeOH and the precipitates were filtered to obtain Intermediate 1-1 (25 g) in a 81% yield.

4) Preparation of Compound 1

(123) After dissolving Compound 1-1 (11 g, 0.0235 mol, 1 eq.) and 2-([1,1'-biphenyl]-4-yl)-4-(4-bromonaphthalen-1-yl)-6-phenyl-1,3,5-triazine (12.8 g, 0.025 mol, 1.05 eq.) in 1,4-dioxane (100 ml) and H₂O (30 ml), Pd(PPh₃)₄ (1.4 g, 0.00118 mol, 0.05 eq.) and K₃PO₄ (15 g, 0.71 mol, 3 eq.) were introduced thereto, and the result was stirred for 4 hours under reflux.

(124) The reaction solution was cooled, then MeOH slurried to precipitate solids, and filtered. The solids were dissolved in 1,2-dichlorobenzene and silica gel filtered. The organic solution was concentrated, MC/MeOH slurried, and then filtered to obtain Compound 1 (12 g) in a 67% yield.

(125) Compound P was synthesized in the same manner as in Preparation Example 1 except that Intermediate A of the following Table 1 was used instead of 2-([1,1'-biphenyl]-4-yl)-4-(4-bromonaphthalen-1-yl)-6-phenyl-1,3,5-triazine.

(126) TABLE-US-00001 TABLE 1 Compound Number Intermediate A Compound P Yield

		67%	4			78%	7
	00		72%	8	01		79%
15	03		04		79%	18	05
	85%	21	07		08		82%
	0		91%	35			78%
		82%	41				78%
		81%	47		0		92%
		88%	53				89%
		78%	61				82%
	0		95%	67			92%
		90%	76				89%
		86%	81		0		96%
		93%	86				88%
		81%	92				92%
	0		89%	101			90%
		92%	107				95%
		87%	113		0		86%
				82%			

<Preparation Example 2> Preparation of Compound 122

(127) ##STR00263## ##STR00264##

1) Preparation of Compound 122-4

(128) Compound 122-4 (38 g, 92% yield) was obtained in the same manner as in Preparation Example 1 except that 2-bromo-5-chloroaniline was used instead of 2,6-dibromoaniline. 2)

Preparation of Compound 122-3

(129) After dissolving Compound 122-4 (38 g, 0.129 mol, 1 eq.) in tetrahydrofuran (THF),

triethylamine (TEA) (0.194 mol) and benzoyl chloride (0.194 mol) were added thereto at 0° C., and then the result was stirred for 1 hour at room temperature. After the reaction was finished, the result was extracted with distilled water and ethyl acetate (EA). The organic layer was dried with MgSO.sub.4, then filtered, and concentrated. The result was adsorbed to silica gel, and purified by column chromatography using dichloromethane and hexane as a developing solvent to obtain Compound 122-3 (48.5 g) in a 95% yield.

3) Preparation of Compound 122-2

(130) After dissolving Compound 122-3 (48.5 g, 0.121 mol, 1 eq.) in nitrobenzene, POCl.sub.3 (0.182 mol) was added thereto, and the result was stirred for 17 hours at 150° C. After the reaction was finished, the result was neutralized at room temperature, and then extracted with distilled water and dichloromethane. The organic layer was dried with MgSO.sub.4, then filtered, and concentrated. The concentrated residue was slurried with MeOH, and then filtered to obtain Compound 122-2 (26 g) in a 56% yield.

4) Preparation of Compound 122-1

(131) After adding 4,4,4',4',5,5,5',5'-octamethyl-2,2'-bi(1,3,2-dioxaborolane) (0.102 mol), Pd(dba).sub.2 (0.0068 mol), XPhos (0.0136 mol), KoAc (0.204 mol) and 1,4-dioxane to Compound 122-2 (26 g, 0.068 mol, 1 eq.), the result was stirred for 5 hours at 110° C. After the reaction was completed, the result was base filtered, and then washed with dichloromethane. The organic solution was concentrated, and then dissolved in a small amount of chloroform, and passed through silica gel. The solution was concentrated, and then MC/MeOH slurried to obtain Compound 122-1 (31 g) in a 97% yield.

5) Preparation of Compound 122

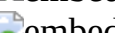
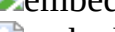
(132) After dissolving Compound 122-1 (10 g, 0.039 mol, 1 eq.) and 2-chloro-4,6-diphenyl-1,3,5-triazine (0.041 mol, 1.05 eq.) in 1,4-dioxane (100 ml) and H.sub.2O (30 ml), Pd(PPh.sub.3).sub.4 (0.0019 mol, 0.05 eq.) and K.sub.3PO.sub.4 (0.117 mol, 3 eq.) were introduced thereto, and the result was stirred for 4 hours under reflux.

(133) The reaction solution was cooled, then MeOH slurried to precipitate solids, and then filtered. The solids were dissolved in 1,2-dichlorobenzene and silica gel filtered. The organic solution was concentrated, MC/MeOH slurried, and then filtered to obtain Compound 122 (20 g) in a 78% yield.

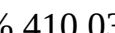
(134) Compound P was synthesized in the same manner as in Preparation Example 2 except that Intermediate A of the following Table 2 was used instead of 2-([1,1'-biphenyl]-4-yl)-4-chloro-6-phenyl-1,3,5-triazine.

(135) TABLE-US-00002

TABLE 2	Compound Number	Intermediate A	Compound P	Yield
122				
124				78%
126				78%
130				72%
135				79%
138				79%
147				69%
149				82%
152				84%
155				78%
161				82%
164				82%
168				81%
172				92%
178				93%
181				95%
186				96%
190				87%
194				83%
198				75%
202				84%
206				79%
210				80%
214				88%
218				89%
222				91%
226				82%
230				83%
234				93%
238				94%
242				89%
246				78%
250				79%

254  75% 258  81%
262  82% 266  91%
270  92% 274  86%
278  87%

(136) Compound P was obtained in the same manner as in Preparation Example 2 except that 4-chloro-2-(dibenzo[b,d]furan-1-yl)aniline was synthesized instead of Compound 122-4 using 2-bromo-4-chloroaniline instead of 2-bromo-5-chloroaniline, and Intermediate A of the following Table 3 was used instead of 2-([1,1'-biphenyl]-4-yl)-4-chloro-6-phenyl-1,3,5-triazine.

(137) TABLE-US-00003 TABLE 3 Com- pound Num- ber Intermediate A Compound P Yield
 82% 286  81% 290
 92% 294  78% 298
 93% 302  72% 306
 79% 310  79% 314
 69% 318  82% 326
 84% 330  78% 334
 82% 338  81% 342
 82% 346  92% 350
 93% 358  88% 366
 91% 370  92% 374
 87% 378  75% 382
 79% 386  81% 390
 92% 394  88% 398
 78% 402  79%
406 01  82% 410 03 
 86% 414 05  89% 418 07
 92% 422 09  93%
426  83% 430  78%
434  72% 438  81%

(138) Compound P was synthesized in the same manner as in Preparation Example 1 except that dibenzo[b,d]thiophen-1-ylboronic acid was used instead of dibenzo[b,d]furan-1-ylboronic acid, and Intermediate A of the following Table 4 was used instead of 2-([1,1'-biphenyl]-4-yl)-4-(4-bromonaphthalen-1-yl)-6-phenyl-1,3,5-triazine.

(139) TABLE-US-00004 TABLE 4 Com- pound Num- ber Intermediate A Compound P Yield
 81% 446  78% 450
 75% 454  81% 458
 78% 462  72% 466
 79% 470  79% 474
 69% 478  82% 482
 84% 486  82% 490
 78% 498  82% 502
 75% 506  82% 510
 72% 514  81% 518
 83% 522  82% 526
 78% 530  72% 534
 81% 538  85% 542
 91% 546  88% 550
 84% 554  92% 558
 87%

(140) Compound P was synthesized in the same manner as in Preparation Example 2 except that 5-chloro-2-(dibenzo[b,d]thiophen-1-yl)aniline was obtained instead of Compound 122-4 using

dibenzo[b,d]thiophen-1-ylboronic acid instead of dibenzo[b,d]furan-1-ylboronic acid, and Intermediate B of the following Table 5 was used instead of 2-chloro-4,6-diphenyl-1,3,5-triazine.

(141) TABLE-US-00005 TABLE 5 Compound Number Intermediate B Compound P Yield 562

82% 566 0 81% 570
 78% 574 78% 578
 72% 582 71% 586
 0 79% 590 79% 594
 69% 598 82% 602
 84% 606 00 78%
610 01 02 82% 614 03 04

71% 618 05 06 82% 622 07
 08 72% 626 09 0 68%
630 91% 634 72%
638 75% 642 89%
646 0 69% 650 75%
654 78% 658 81%
662 91% 666 0 89%
670 82% 674 81%
678 78% 682 77%
686 0 82% 690 91%
694 68% 698 71%

(142) Compound P was obtained in the same manner as in Preparation Example 2 except that 4-chloro-2-(dibenzo[b,d]thiophen-1-yl) aniline was synthesized using dibenzo[b,d]thiophen-1-ylboronic acid instead of dibenzo [b,d]furan-1-ylboronic acid and 2-bromo-4-chloroaniline instead of 2-bromo-5-chloroaniline, and Intermediate A of the following Table 6 was used instead of 2-chloro-4,6-diphenyl-1,3,5-triazine.

(143) TABLE-US-00006 TABLE 6 Compound Number Intermediate A Compound P Yield 702

81% 706 0 72% 710
 78% 714 72% 718
 86% 722 79% 726
 0 79% 730 69% 734
 82% 742 78% 750
 69% 754 0 82% 758
 82% 762 68% 766
 71% 770 72% 774
 0 82% 778 89% 782
 82% 786 89% 790
 82% 794 0 83% 798
 78% 802 79% 806
 69% 810 71% 814
 00 77% 818 01 02 82% 822 03 04 85% 826 05 06
 69% 830 07 08 67%

(144) Compounds were prepared in the same manner as in the preparation examples, and the synthesis identification results are shown in the following Table 7.

(145) TABLE-US-00007 TABLE 7 Compound FD-MS 1 m/z = 778.90 (C₅₆H₃₄N₄O = 778.27) 4 m/z = 571.66 (C₄₃H₂₅NO = 571.19) 8 m/z = 521.61 (C₃₉H₂₃NO = 521.18) 12 m/z = 597.70 (C₄₅H₂₇NO = 597.21) 15 m/z = 628.76 (C₄₆H₃₂N₂O = 628.25) 18 m/z = 599.68 (C₄₃H₂₅N₃O = 599.20) 21 m/z = 725.83 (C₅₃H₃₁N₃O = 725.25) 25 m/z = 736.86 (C₅₅H₃₂N₂O = 736.25) 28 m/z = 702.80 (C₅₁H₃₀N₂O₂ = 702.23) 32 m/z = 676.76 (C₄₈H₂₈N₄O = 676.23) 35 m/z = 652.74

(C46H28N4O = 652.23) 38 m/z = 742.82 (C52H30N4O2 = 742.24) 41 m/z = 742.82
(C52H30N4O2 = 742.24) 44 m/z = 732.85 (C50H28N4OS = 732.20) 47 m/z = 768.90
(C55H36N4O = 768.29) 50 m/z = 728.84 (C52H32N4O = 728.2) 53 m/z = 867.99 (C62H37N5O =
867.30) 58 m/z = 884.05 (C63H37N3OS = 883.27) 61 m/z = 697.76 (C49H32NO2P = 697.22) 63
m/z = 597.70 (C45H27NO = 597.21) 67 m/z = 785.93 (C60H35NO = 785.27) 73 m/z = 651.75
(C47H29N3O = 651.23) 76 m/z = 665.74 (C47H27N3O2 = 665.21) 78 m/z = 741.83
(C53H31N3O2 = 741.24) 81 m/z = 665.74 (C47H27N3O2 = 665.21) 83 m/z = 681.80
(C47H27N3OS = 681.19) 86 m/z = 816.94 (C59H36N4O = 816.29) 90 m/z = 664.75
(C47H28N4O = 664.23) 92 m/z = 727.85 (C53H33N3O = 727.26) 95 m/z = 575.66 (C41H25N3O
= 575.20) 101 m/z = 651.75 (C47H29N3O = 651.23) 103 m/z = 803.94 (C59H37N3O = 803.29)
107 m/z = 665.74 (C47H27N3O2 = 665.21) 110 m/z = 681.80 (C47H27N3OS = 681.19) 113 m/z =
741.83 (C53H31N3O2 = 741.24) 115 m/z = 816.94 (C59H36N4O = 816.29) 122 m/z = 576.64
(C40H24N4O = 576.20) 124 m/z = 728.84 (C52H32N4O = 728.26) 126 m/z = 652.74
(C46H28N4O = 652.23) 130 m/z = 666.72 (C46H26N4O2 = 666.21) 135 m/z = 742.82
(C52H30N4O2 = 742.24) 138 m/z = 817.93 (C58H35N5O = 817.28) 142 m/z = 692.80
(C49H32N4O = 692.26) 144 m/z = 804.93 (C58H36N4O = 804.29) 147 m/z = 521.61 (C39H23NO
= 521.18) 149 m/z = 571.66 (C43H25NO = 571.19) 152 m/z = 649.78 (C49H31NO = 649.24) 155
m/z = 548.63 (C40H24N2O = 548.19) 161 m/z = 573.68 (C43H27NO = 573.21) 164 m/z = 603.73
(C43H25NOS = 603.17) 168 m/z = 752.90 (C56H36N2O = 752.28) 172 m/z = 652.74
(C46H28N4O = 652.23) 174 m/z = 634.72 (C47H26N2O = 634.20) 178 m/z = 751.87
(C55H33N3O = 751.26) 181 m/z = 758.89 (C52H30N4OS = 758.21) 186 m/z = 742.82
(C52H30N4O2 = 742.24) 190 m/z = 742.82 (C52H30N4O2 = 742.24) 194 m/z = 732.85
(C50H28N4OS = 732.20) 198 m/z = 690.75 (C47H26N6O = 690.22) 201 m/z = 618.74
(C43H26N2OS = 618.18) 202 m/z = 660.76 (C49H28N2O = 660.22) 206 m/z = 741.83
(C53H31N3O2 = 741.24) 210 m/z = 712.83 (C53H32N2 = 712.25) 214 m/z = 649.78 (C49H31NO
= 649.24) 218 m/z = 727.85 (C53H33N3O = 727.26) 221 m/z = 665.74 (C47H27N3O2 = 665.21)
222 m/z = 665.74 (C47H27N3O2 = 665.21) 226 m/z = 741.83 (C53H31N3O2 = 741.24) 230 m/z =
816.94 (C59H36N4O = 816.29) 234 m/z = 740.85 (C53H32N4O = 740.26) 238 m/z = 651.75
(C47H29N3O = 651.23) 241 m/z = 727.85 (C53H33N3O = 727.26) 242 m/z = 727.85
(C53H33N3O = 727.26) 246 m/z = 665.74 (C47H27N3O2 = 665.21) 250 m/z = 741.83
(C53H31N3O2 = 741.24) 254 m/z = 816.94 (C59H36N4O = 816.29) 258 m/z = 740.85
(C53H32N4O = 740.26) 261 m/z = 651.75 (C47H29N3O = 651.23) 262 m/z = 665.74
(C47H27N3O2 = 665.21) 266 m/z = 681.80 (C47H27N3OS = 681.19) 270 m/z = 791.89
(C57H33N3O2 = 791.26) 274 m/z = 816.94 (C59H36N4O = 816.29) 278 m/z = 691.82
(C50H33N3O = 691.26) 280 m/z = 652.74 (C46H28N4O = 652.23) 282 m/z = 728.84
(C52H32N4O = 728.26) 286 m/z = 728.84 (C52H32N4O = 728.26) 290 m/z = 666.72
(C46H26N4O2 = 666.21) 294 m/z = 742.82 (C52H30N4O2 = 742.24) 298 m/z = 741.84
(C52H31N5O = 741.25) 301 m/z = 778.90 (C56H34N4O = 778.27) 302 m/z = 728.84
(C52H32N4O = 728.26) 306 m/z = 741.84 (C52H31N5O = 741.25) 310 m/z = 521.61 (C39H23NO
= 521.18) 314 m/z = 597.70 (C45H27NO = 597.21) 318 m/z = 548.63 (C40H24N2O = 548.19) 321
m/z = 527.63 (C37H21NOS = 527.13) 322 m/z = 552.66 (C40H28N2O = 552.22) 326 m/z =
663.76 (C49H29NO2 = 663.22) 330 m/z = 662.78 (C49H30N2O = 662.24) 334 m/z = 652.74
(C46H28N4O = 652.23) 338 m/z = 549.62 (C39H23N3O = 549.18) 341 m/z = 571.66 (C43H25NO
= 571.19) 342 m/z = 521.61 (C39H23NO = 521.18) 346 m/z = 597.70 (C45H27NO = 597.21) 350
m/z = 548.63 (C40H24N2O = 548.19) 354 m/z = 676.80 (C50H32N2O = 676.25) 358 m/z =
751.87 (C55H33N3O = 751.26) 361 m/z = 602.68 (C43H26N2O2 = 602.20) 362 m/z = 742.88
(C53H30N2OS = 742.21) 366 m/z = 728.84 (C52H32N4O = 728.26) 370 m/z = 742.82
(C52H30N4O2 = 742.24) 374 m/z = 742.82 (C52H30N4O2 = 742.24) 378 m/z = 867.99
(C62H37N5O = 867.30) 381 m/z = 628.76 (C46H32N2O = 628.25) 382 m/z = 602.68
(C43H26N2O2 = 602.20) 386 m/z = 675.77 (C49H29N3O = 675.23) 390 m/z = 713.82

(C53H31NO₂ = 713.24) 394 m/z = 662.78 (C49H30N₂O = 662.24) 398 m/z = 649.78 (C49H31NO = 649.24) 401 m/z = 727.85 (C53H33N₃O = 727.26) 402 m/z = 727.85 (C53H33N₃O = 727.26) 406 m/z = 665.74 (C47H27N₃O₂ = 665.21) 410 m/z = 741.83 (C53H31N₃O₂ = 741.24) 414 m/z = 816.94 (C59H36N₄O = 816.29) 418 m/z = 664.75 (C47H28N₄O = 664.23) 421 m/z = 803.94 (C59H37N₃O = 803.29) 422 m/z = 681.80 (C47H27N₃OS = 681.19) 426 m/z = 665.74 (C47H27N₃O₂ = 665.21) 430 m/z = 741.83 (C53H31N₃O₂ = 741.24) 434 m/z = 757.90 (C53H31N₃OS = 757.22) 438 m/z = 664.75 (C47H28N₄O = 664.23) 441 m/z = 744.90 (C52H32N₄S = 744.23) 442 m/z = 682.79 (C46H26N₄OS = 682.18) 446 m/z = 682.79 (C46H26N₄OS = 682.18) 450 m/z = 757.90 (C52H31N₅S = 757.23) 454 m/z = 784.97 (C55H36N₄S = 784.27) 458 m/z = 758.89 (C52H30N₄OS = 758.21) 461 m/z = 587.73 (C43H25NS = 587.17) 462 m/z = 537.67 (C39H23NS = 537.16) 466 m/z = 613.77 (C45H27NS = 613.19) 470 m/z = 564.70 (C40H24N₂S = 564.17) 474 m/z = 602.74 (C43H26N₂S = 602.18) 478 m/z = 675.84 (C50H29NS = 675.20) 480 m/z = 644.82 (C46H32N₂S = 644.23) 481 m/z = 543.70 (C37H21NS₂ = 543.11) 482 m/z = 543.70 (C37H21NS₂ = 543.11) 486 m/z = 758.89 (C52H30N₄OS = 758.21) 490 m/z = 615.74 (C43H25N₃S = 615.18) 494 m/z = 845.02 (C60H36N₄S = 844.27) 498 m/z = 758.89 (C52H30N₄OS = 758.21) 501 m/z = 758.89 (C52H30N₄OS = 758.21) 502 m/z = 884.06 (C62H37N₅S = 883.28) 506 m/z = 821.00 (C58H36N₄S = 820.27) 510 m/z = 744.90 (C52H32N₄S = 744.23) 514 m/z = 774.95 (C52H30N₄S₂ = 774.19) 518 m/z = 850.06 (C59H35N₃S₂ = 849.23) 521 m/z = 697.87 (C47H27N₃S₂ = 697.16) 522 m/z = 756.91 (C53H32N₄S = 756.23) 526 m/z = 743.91 (C53H33N₃S = 743.24) 530 m/z = 743.91 (C53H33N₃S = 743.24) 534 m/z = 833.01 (C59H36N₄S = 832.27) 538 m/z = 681.80 (C47H27N₃OS = 681.19) 541 m/z = 870.07 (C63H39N₃S = 869.29) 542 m/z = 591.72 (C41H25N₃S = 591.18) 546 m/z = 757.90 (C53H31N₃OS = 757.22) 550 m/z = 757.90 (C53H31N₃OS = 757.22) 554 m/z = 668.81 (C46H28N₄S = 668.20) 558 m/z = 773.96 (C53H31N₃S₂ = 773.20) 561 m/z = 682.79 (C46H26N₄OS = 682.18) 562 m/z = 682.79 (C46H26N₄OS = 682.18) 566 m/z = 758.89 (C52H30N₄OS = 758.21) 570 m/z = 774.95 (C52H30N₄S₂ = 774.19) 574 m/z = 681.81 (C46H27N₅S = 681.20) 578 m/z = 587.73 (C43H25NS = 587.17) 581 m/z = 589.75 (C43H27NS = 589.19) 582 m/z = 543.70 (C37H21NS₂ = 543.11) 586 m/z = 668.81 (C46H28N₄S = 668.20) 590 m/z = 834.98 (C58H34N₄OS = 834.25) 594 m/z = 564.70 (C40H24N₂S = 564.17) 598 m/z = 602.74 (C43H26N₂S = 602.18) 601 m/z = 568.73 (C40H28N₂S = 568.20) 602 m/z = 644.82 (C46H32N₂S = 644.23) 606 m/z = 675.84 (C50H29NS = 675.20) 610 m/z = 869.04 (C62H36N₄S = 868.27) 614 m/z = 884.06 (C62H37N₅S = 883.28) 618 m/z = 834.00 (C58H35N₅S = 833.26) 621 m/z = 744.90 (C52H32N₄S = 744.23) 622 m/z = 821.00 (C58H36N₄S = 820.27) 626 m/z = 758.89 (C52H30N₄OS = 758.21) 630 m/z = 901.11 (C62H36N₄S₂ = 900.24) 634 m/z = 758.89 (C52H30N₄OS = 758.21) 638 m/z = 678.84 (C49H30N₂S = 678.21) 641 m/z = 644.82 (C46H32N₂S = 644.23) 642 m/z = 618.74 (C43H26N₂OS = 618.18) 646 m/z = 691.84 (C49H29N₃S = 691.21) 650 m/z = 613.77 (C45H27NS = 613.19) 654 m/z = 697.87 (C47H27N₃S₂ = 697.16) 658 m/z = 773.96 (C53H31N₃S₂ = 773.20) 661 m/z = 591.72 (C41H25N₃S = 591.18) 662 m/z = 667.82 (C47H29N₃S = 667.21) 666 m/z = 743.91 (C53H33N₃S = 743.24) 670 m/z = 681.80 (C47H27N₃OS = 681.19) 674 m/z = 681.80 (C47H27N₃OS = 681.19) 678 m/z = 756.91 (C53H32N₄S = 756.23) 681 m/z = 820.01 (C59H37N₃S = 819.27) 682 m/z = 667.82 (C47H29N₃S = 667.21) 686 m/z = 681.80 (C47H27N₃OS = 681.19) 670 m/z = 757.90 (C53H31N₃OS = 757.22) 674 m/z = 743.91 (C53H33N₃S = 743.24) 678 m/z = 783.98 (C56H37N₃S = 783.27) 681 m/z = 820.01 (C59H37N₃S = 819.27) 682 m/z = 667.82 (C47H29N₃S = 667.21) 686 m/z = 681.80 (C47H27N₃OS = 681.19) 690 m/z = 757.90 (C53H31N₃OS = 757.22) 694 m/z = 743.91 (C53H33N₃S = 743.24) 698 m/z = 783.98 (C56H37N₃S = 783.27) 701 m/z = 592.71 (C40H24N₄S = 592.17) 702 m/z = 668.81 (C46H28N₄S = 668.20) 706 m/z = 682.79 (C46H26N₄OS = 682.18) 710 m/z = 758.89 (C52H30N₄OS = 758.21) 714 m/z = 834.00 (C58H35N₅S = 833.26) 718 m/z = 681.81 (C46H27N₅S = 681.20) 721 m/z = 613.77 (C45H27NS = 613.19) 722 m/z = 665.84

(C49H31N3 = 665.22) 726 m/z = 539.15 (C37H21N3S = 539.15) 730 m/z = 564.70 (C40H24N2S = 564.17) 734 m/z = 678.84 (C49H30N2S = 678.21) 738 m/z = 692.87 (C50H32N2S = 692.23) 741 m/z = 676.83 (C49H28N2S = 676.20) 742 m/z = 768.92 (C54H32N4S = 768.23) 746 m/z = 758.89 (C52H30N4OS = 758.21) 750 m/z = 748.91 (C50H28N4S2 = 748.18) 754 m/z = 758.89 (C52H30N4OS = 758.21) 758 m/z = 834.00 (C58H35N5S = 833.26) 761 m/z = 706.81 (C47H26N6S = 706.19) 762 m/z = 644.82 (C46H32N2S = 644.23) 766 m/z = 757.90 (C53H31N3OS = 757.22) 770 m/z = 678.84 (C49H30N2S = 678.21) 774 m/z = 619.80 (C43H25NS2 = 619.14) 778 m/z = 678.84 (C49H30N2S = 678.21) 781 m/z = 591.72 (C41H25N3S = 591.18) 782 m/z = 667.82 (C47H29N3S = 667.21) 786 m/z = 820.01 (C59H37N3S = 819.27) 790 m/z = 681.80 (C47H27N3OS = 681.19) 794 m/z = 757.90 (C53H31N3OS = 757.22) 798 m/z = 833.01 (C59H36N4S = 832.27) 801 m/z = 667.82 (C47H29N3S = 667.21) 802 m/z = 667.82 (C47H29N3S = 667.21) 806 m/z = 681.80 (C47H27N3OS = 681.19) 810 m/z = 668.81 (C46H28N4S = 668.20) 814 m/z = 743.91 (C53H33N3S = 743.24) 818 m/z = 743.91 (C53H33N3S = 743.24) 821 m/z = 757.90 (C53H31N3OS = 757.22) 823 m/z = 833.01 (C59H36N4S = 832.27) 826 m/z = 707.88 (C50H33N3S = 707.24) 828 m/z = 757.90 (C53H31N3OS = 757.22) 830 m/z = 783.98 (C56H37N3S = 783.27) 831 m/z = 756.91 (C53H32N4S = 756.23)

(146) TABLE-US-00008 TABLE 8 NO .sup.1H NMR (CDCl.sub.3, 300 Mz) 1 δ = 8.55(t, 2H), 8.30~8.21(m, 5H), 8.02~8.01(m, 3H), 7.85~7.90(m, 4H), 7.66(m, 2H), 7.54~7.25(m, 18H) 4 δ = 9.15(s, 1H), 8.93(t, 2H), 8.30~8.02(m, 8H), 7.92~7.82(m, 6H), 7.66(m, 2H), 7.54~7.32(m, 6H) 8 δ = 8.93(m, 2H), 8.30(m, 2H), 8.21(m, 1H), 8.12(m, 2H), 8.02(t, 1H), 7.93~7.82(m, 7H), 7.66(m, 2H), 7.54~7.32(m, 6H) 12 δ = 8.93(t, 2H), 8.30(t, 2H), 8.21(d, 2H), 8.12(t, 2H), 8.02(d, 1H), 7.92~7.82(m, 7H), 7.66(m, 2H), 7.54~7.25(m, 10H) 15 δ = 8.30(t, 2H), 8.21(d, 1H), 8.02(d, 1H), 7.92~7.89(m, 2H), 7.66(m, 2H), 7.54~7.32(m, 8H), 7.05~7.02(m, 4H), 6.73~6.69(m, 4H), 6.55(m, 2H), 1.72(s, 2H) 18 δ = 8.88(m, 1H), 8.30(m, 4H), 8.10~8.06(m, 4H), 7.89~7.92(m, 2H), 7.81(m, 1H), 7.69~7.66(m, 2H), 7.54~7.32(m, 11H) 21 δ = 8.55(m, 3H), 8.30(m, 4H), 8.21(d, 1H), 8.10~8.02(m, 5H), 7.92~7.89(m, 2H), 7.81(m, 1H), 7.66(m, 2H), 7.55~7.32(m, 13H) 25 δ = 9.66(d, 1H), 8.93(t, 2H), 8.55(t, 2H), 8.21~8.12(m, 5H), 7.89~7.79(m, 9H), 7.68~7.51(m, 8H), 7.38~7.29(m, 5H) 28 δ = 8.33(d, 1H), 8.21(d, 1H), 8.02(m, 1H), 7.91~7.89(m, 6H), 7.66(m, 2H), 7.54~7.50(m, 3H), 7.39~7.32(m, 6H), 6.92~6.89(m, 4H), 6.79~6.69(m, 4H), 6.59(m, 2H) 32 δ = 8.49(t, 2H), 8.30(t, 2H), 8.21(d, 1H), 8.02~7.89(m, 9H), 7.66~7.32(m, 12H) 35 δ = 8.30~8.21(m, 8H), 8.02(d, 1H), 7.92~7.89(m, 2H), 7.66~7.32(m, 17H) 38 δ = 8.30~8.21(m, 6H), 8.02(d, 1H), 7.92~7.81(m, 5H), 7.70~7.32(m, 18H), 41 δ = 8.30(m, 2H), 8.21(d, 1H), 8.02(d, 1H), 7.92~7.85(m, 5H), 7.75(m, 1H), 7.66~7.32(m, 20H) 44 δ = 9.09(d, 1H), 8.49~8.45(t, 2H), 8.30(t, 2H), 8.21(m, 1H), 8.02~7.92(m, 9H), 7.66~7.32(m, 13H) 47 δ = 8.30~8.21(m, 7H), 8.02(d, 1H), 7.92~7.85(m, 4H), 7.66~7.25(m, 20H) 50 δ = 8.30~8.21(m, 8H), 8.02(d, 1H), 7.92~7.89(m, 2H), 7.66~7.32(m, 21H) 53 δ = 8.55(d, 1H), 8.30~8.21(m, 4H), 8.00~7.89(m, 8H), 7.59~7.25(m, 24H) 58 δ = 8.55(m, 1H), 8.41(d, 1H), 8.30~8.20(m, 7H), 8.11~8.02(m, 4H), 7.89~7.76(m, 7H), 7.54~7.32(m, 17H) 61 δ = 8.30(t, 2H), 8.21(d, 1H), 8.02(d, 1H), 7.79~7.32(m, 28H) 63 δ = 8.93(t, 3H), 8.30(t, 2H), 8.21(m, 1H), 8.12~7.54(m, 2H), 8.02(d, 1H), 7.93~7.82(m, 7H), 7.57~7.38(m, 12H) 67 δ = 8.40(t, 2H), 8.30(t, 2H), 8.21(m, 1H), 8.00~7.75(m, 9H), 7.54~7.16(m, 21H) 73 δ = 8.30~8.21(m, 8H), 8.02(d, 1H), 7.92~7.89(m, 2H), 7.79(m, 2H), 7.66(m, 2H), 7.54~7.32(m, 14H) 76 δ = 8.37~8.28(m, 6H), 8.02(d, 1H), 7.92~7.85(m, 4H), 7.66~7.32(m, 16H) 78 δ = 8.30~8.21(m, 8H), 8.02(d, 1H), 7.92~7.85(m, 4H), 7.66~7.32(m, 18H) 81 δ = 8.37~8.28(m, 6H), 8.02~7.89(m, 5H), 7.66~7.75(m, 5H), 7.54~7.32(m, 11H) 83 δ = 8.45(d, 1H), 8.37~8.28(m, 6H), 8.02~7.82(m, 6H), 7.66(t, 2H), 7.54~7.32(m, 12H) 86 δ = 8.55(d, 1H), 8.30~8.21(m, 8H), 8.02(d, 1H), 7.94~7.87(m, 4H), 7.66~7.25(m, 21H) 90 δ = 8.55(d, 1H), 8.37~8.28(m, 5H), 8.12(d, 1H), 8.02(m, 1H), 7.94~7.89(m, 3H), 7.66~7.32(m, 17H) 92 δ = 8.30~8.23(m, 8H), 8.02(d, 1H), 7.92~7.79(m, 6H), 7.66(m, 2H), 7.51~7.25(m, 16H) 95 δ = 8.30(m, 2H), 8.23~8.21(m, 2H), 8.02(m, 1H), 7.92~7.89(m, 2H), 7.79(m, 3H), 7.66(m, 2H), 7.54~7.32(m, 12H) 101 δ = 8.30~8.21(m, 8H), 8.02(d, 1H), 7.92~7.89(m, 2H), 7.79(t, 2H), 7.66(m, 2H), 7.54~7.32(m, 14H) 103 δ = 8.30~8.21(m, 5H), 8.02(d,

1H), 7.92~7.89(m, 2H), 7.79~7.25(m, 28H) 107 δ = 8.30(m, 2H), 8.02~7.89(m, 5H), 7.79~7.66(m, 7H), 7.51~7.32(m, 11H) 110 δ = 8.45(d, 1H), 8.30(m, 2H), 8.23~8.21(m, 2H), 7.98~7.79(m, 8H), 7.66(m, 2H), 7.52~7.32(m, 12H) 113 δ = 8.30~8.21(m, 6H), 8.02~7.89(m, 5H), 7.75~7.64(m, 7H), 7.54~7.32(m, 13H) 115 δ = 8.55(d, 1H), 8.30~8.23(m, 6H), 8.02(d, 1H), 7.94~7.87(m, 4H), 7.79~7.77(m, 3H), 7.69~7.32(m, 21H) 122 δ = 8.30~8.28(m, 7H), 8.12(d, 1H), 8.30(d, 1H), 7.92~7.89(m, 2H), 7.66(m, 1H), 7.54~7.32(m, 12H) 124 δ = 8.30~8.27(m, 3H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.85(m, 6H), 7.66(d, 1H), 7.52~7.25(m, 20H) 126 δ = 8.30~8.24(m, 6H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.89(m, 2H), 7.70~7.66(m, 2H), 7.54~7.32(m, 16H) 130 δ = 8.30~8.27(m, 5H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.81(m, 5H), 7.66(m, 2H), 7.54~7.32(m, 12H) 135 δ = 8.30~8.27(m, 5H), 8.12(d, 1H), 8.03(d, 1H), 7.95~7.85(m, 5H), 7.75(d, 1H), 7.66~7.64(m, 3H), 7.54~7.25(m, 13H) 138 δ = 8.55(d, 1H), 8.30~8.27(m, 5H), 8.12(d, 1H), 8.03(m, 1H), 7.94~7.85(m, 6H), 7.77(d, 1H), 7.69~7.33(m, 20H) 142 δ = 8.30~8.28(m, 5H), 8.12(m, 1H), 8.06~8.03(m, 2H), 7.92~7.87(m, 3H), 7.54~7.28(m, 25H) 144 δ = 8.30~8.28(m, 7H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.85(m, 4H), 7.66(m, 1H), 7.54~7.25(m, 22H) 147 δ = 9.15(d, 1H), 8.93(d, 1H), 8.30~8.27(m, 3H), 8.18~8.03(m, 5H), 7.89~7.82(m, 4H), 7.71~7.66(m, 3H), 7.54~7.32(m, 6H) 149 δ = 9.15(d, 1H), 8.93(m, 2H), 8.30~8.27(m, 3H), 8.18~8.03(m, 6H), 7.92~7.82(m, 6H), 7.66(d, 1H), 7.54~7.32(m, 6H) 152 δ = 8.30~8.27(m, 3H), 8.12(d, 1H), 7.03(d, 1H), 7.92~7.89(m, 2H), 7.66(m, 4H), 7.54~7.32(m, 20H) 155 δ = 8.88(d, 1H), 8.30~8.27(m, 5H), 8.12~8.03(m, 3H), 7.89~7.82(m, 4H), 7.66(m, 2H), 7.54~7.32(m, 9H) 161 δ = 8.30~8.27(m, 3H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.89(m, 2H), 7.66(m, 4H), 7.54~7.41(m, 16H) 164 δ = 8.45~8.41(m, 2H), 8.30(m, 2H), 8.20(m, 1H), 8.72(m, 1H), 8.03~7.89(m, 3H), 7.58~7.25(m, 14H) 168 δ = 8.30~8.27(m, 3H), 8.12(m, 1H), 8.03(m, 1H), 7.92~7.89(m, 2H), 7.66(m, 1H), 7.54~7.26(m, 14H), 7.11~6.98(m, 8H), 6.69(m, 4H), 6.51(t, 2H) 172 δ = 8.30~8.24(m, 8H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.89(t, 2H), 7.70~7.66(m, 2H), 7.57~7.32(m, 14H) 174 δ = 8.85(d, 1H), 8.38(d, 1H), 8.27(d, 1H), 8.12(m, 3H), 8.03~7.03(m, 6H), 7.66~7.50(m, 10H), 7.92~7.29(m, 4H) 178 δ = 8.55(m, 2H), 8.30~8.27(m, 3H), 8.12~8.03(m, 6H), 7.94~7.89(m, 4H), 7.66~7.25(m, 18H) 181 δ = 8.45(d, 1H), 8.30~8.24(m, 6H), 8.12(d, 1H), 8.03~7.98(m, 6H), 7.70~7.66(t, 2H), 7.57~7.32(m, 14H) 186 δ = 8.30~8.24(m, 4H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.81(m, 5H), 7.70~7.66(m, 3H), 7.57~7.32(m, 16H) 190 δ = 8.30~8.24(m, 6H), 8.12(m, 1H), 8.03(d, 1H), 7.92~7.85(m, 5H), 7.70~7.32(m, 17H) 194 δ = 9.09(d, 1H), 8.49~8.45(m, 2H), 8.30~8.27(m, 3H), 8.12(m, 1H), 8.00~7.89(m, 9H), 7.59~7.32(m, 12H) 198 δ = 8.56~8.55(m, 2H), 8.30~8.27(m, 3H), 8.12(m, 2H), 8.03(d, 1H), 7.94~7.88(m, 4H), 7.69~7.25(m, 14H) 201 δ = 8.30~8.27(m, 3H), 8.12(d, 1H), 8.03(d, 2H), 7.92~7.89(m, 2H), 7.66(m, 1H), 7.54~7.32(m, 7H), 7.21~7.16(m, 6H), 6.97~6.88(m, 4H), 6.59(d, 1H) 202 δ = 8.30~8.27(m, 3H), 8.12(m, 3H), 8.03(d, 1H), 7.92~7.89(m, 2H), 7.70~7.29(m, 19H) 206 δ = 8.27~8.23(m, 6H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.79(m, 7H), 7.66~7.32(m, 16H) 210 δ = 8.49(d, 1H), 8.30~8.27(m, 3H), 8.12~8.03(m, 7H), 7.92~7.89(m, 2H), 7.66~7.29(m, 18H) 214 δ = 8.30~8.27(m, 3H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.89(m, 2H), 7.70~7.66(m, 5H), 7.54~7.32(m, 19H) 218 δ = 8.30~8.24(m, 7H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.79(m, 6H), 7.70~7.66(m, 2H), 7.57~7.32(m, 16H) 221 δ = 8.30~8.23(m, 6H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.89(m, 3H), 7.75(d, 1H), 7.66~7.32(m, 15H) 222 δ = 8.30~8.23(m, 6H), 8.12(d, 1H), 8.03(m, 1H), 7.92~7.85(m, 4H), 7.66~7.32(m, 25H) 226 δ = 8.30~8.23(m, 8H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.85(m, 6H), 7.66~7.32(m, 15H) 230 δ = 8.55(d, 1H), 8.30~8.28(m, 8H), 8.12(d, 1H), 8.03(d, 1H), 7.94~7.85(m, 6H), 7.77(d, 1H), 7.69~7.25(m, 18H) 234 δ = 8.55(d, 1H), 8.30~8.23(m, 6H), 8.12(d, 1H), 8.03(m, 1H), 7.94~7.87(m, 4H), 7.77(d, 1H), 7.69~7.25(m, 18H) 238 δ = 8.30~8.23(m, 8H), 8.12(d, 1H), 8.03(d, 1H), 7.89~7.85(m, 4H), 7.66(d, 1H), 7.54~7.32(m, 14H) 241 δ = 8.30~8.23(m, 5H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.89(m, 2H), 7.75~7.66(m, 4H), 7.57~7.32(m, 20H) 242 δ = 8.30~8.23(m, 8H), 8.12(d, 1H), 8.03(m, 1H), 7.92~7.79(m, 6H), 7.66(d, 1H), 7.54~7.25(m, 16H) 246 δ = 8.30~8.23(m, 6H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.85(m, 4H), 7.66~7.32(m, 15H) 250 δ = 8.30~8.28(m, 8H), 8.12(d, 1H), 8.03(m, 1H), 7.89~7.85(m, 6H), 7.66~7.32(m, 15H) 254 δ = 8.55(d, 1H), 8.30~8.23(m, 8H), 8.12(d, 1H), 8.03(d, 1H), 7.94~7.85(m, 6H), 7.77(d, 1H),

7.66~7.25(m, 18H) 258 δ = 8.55(d, 1H), 8.30~8.28(m, 6H), 8.12(d, 1H), 8.03(d, 1H), 7.94~7.87(m, 4H), 7.77(d, 1H), 7.58~7.25(m, 18H) 261 δ = 8.30~8.23(m, 4H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.79(m, 8H), 7.66(m, 1H), 7.54~7.32(m, 14H) 262 δ = 8.30~8.23(m, 4H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.79(m, 6H), 7.66~7.32(m, 15H) 266 δ = 8.45(d, 1H), 8.30~8.23(m, 4H), 8.12~7.79(m, 10H), 7.66(m, 1H), 7.54~7.32(m, 11H) 270 δ = 2.34~8.23(m, 5H), 8.12(d, 1H), 8.03(d, 1H), 7.92~7.32(m, 13H), 7.54~7.32(m, 12H) 274 δ = 8.55(d, 1H), 8.30~8.23(m, 6H), 8.12(d, 1H), 8.03(d, 1H), 7.94~7.79(m, 9H), 7.66~7.25(m, 18H) 278 δ = 8.30~8.23(m, 4H), 8.12(d, 1H), 8.06~8.03(m, 2H), 7.92~7.87(m, 3H), 7.79(m, 2H), 7.66~7.28(m, 16H), 1.72(s, 2H) 280 δ = 8.30~8.28(m, 7H), 8.04(d, 1H), 7.92~7.89(m, 3H), 7.66(d, 1H), 7.54~7.41(m, 12H) 282 δ = 8.30(m, 2H), 8.21(d, 1H), 8.04(d, 1H), 7.92~7.85(m, 6H), 7.56(d, 1H), 7.54~7.25(m, 20H) 286 δ = 8.55(m, 2H), 8.45(d, 1H), 8.30~8.21(m, 5H), 8.01~7.82(m, 10H), 7.66(d, 1H), 7.55~7.41(m, 13H) 290 δ = 8.30~8.21(m, 5H), 8.04(d, 1H), 7.92~7.85(m, 7H), 7.66(d, 1H), 7.54~7.25(m, 18H) 294 δ = 8.30~8.21(m, 5H), 8.04(d, 1H), 7.90~7.81(m, 8H), 7.66(m, 2H), 7.54~7.25(m, 14H) 298 δ = 8.55(d, 1H), 8.30~8.21(m, 5H), 8.04(m, 1H), 7.94~7.87(m, 5H), 7.77(s, 1H), 7.66~7.25(m, 18H) 301 δ = 8.55(d, 1H), 8.42(d, 1H), 8.28~8.21(m, 5H), 8.08~8.04(m, 3H), 7.92~7.79(m, 7H), 7.66~7.41(m, 17H) 302 δ = 8.28~8.21(m, 6H), 8.04(d, 1H), 7.92~7.85(m, 5H), 7.70~7.66(m, 2H), 7.51~7.25(m, 18H) 306 δ = 8.55(d, 1H), 8.30~8.21(m, 5H), 8.12(d, 1H), 8.04(d, 1H), 7.94~7.85(m, 6H), 7.66~7.25(m, 17H) 310 δ = 8.30(m, 2H), 8.21~8.04(m, 5H), 7.92~7.82(m, 5H), 7.71~7.66(m, 3H), 7.54~7.32(m, 6H) 314 δ = 8.93(m, 2H), 8.30(m, 2H), 8.21(d, 1H), 8.12~8.04(m, 2H), 7.92~7.82(m, 8H), 7.66(m, 1H), 7.54~7.25(m, 10H) 318 δ = 8.30~8.21(m, 5H), 8.09~8.04(m, 2H), 7.92~7.82(m, 5H), 7.66(m, 2H), 7.54~7.32(m, 10H) 321 δ = 8.45(m, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.11~7.98(m, 8H) 322 δ = 8.30(m, 2H), 7.92~7.82(m, 3H), 7.66(m, 1H), 7.54~7.02(m, 6H), 7.32~7.02(m, 4H), 6.80~6.73(m, 3H), 6.55(m, 2H), 1.72(m, 6H) 326 δ = 8.30(m, 2H), 8.21(m, 2H), 8.04(m, 1H), 7.92~7.81(m, 6H), 7.72~7.66(m, 3H), 7.54~7.25(m, 16H) 330 δ = 8.49(m, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.12~8.04(m, 3H), 7.92~7.89(m, 3H), 7.58~7.25(m, 20H) 334 δ = 8.53(m, 2H), 8.30(m, 2H), 8.21(m, 1H), 8.04(m, 1H), 7.92~7.89(m, 3H), 7.70~7.66(m, 3H), 7.54~7.25(m, 10H), 7.14(m, 2H) 338 δ = 8.30~8.16(m, 6H), 8.04(m, 1H), 7.92~7.83(m, 5H), 7.66~7.41(m, 11H) 341 δ = 9.15(d, 1H), 8.92(m, 2H), 8.30(m, 2H), 8.21~8.04(m, 6H), 7.90~7.82(m, 7H), 7.66(m, 1H), 7.54~7.38(m, 6H) 342 δ = 9.15(d, 1H), 8.93(m, 1H), 8.30(m, 2H), 8.21~8.04(m, 5H), 7.92~7.82(m, 5H), 7.66~7.71(m, 3H), 7.54~7.32(m, 6H) 346 δ = 8.93(m, 2H), 8.30(m, 2H), 8.21(m, 1H), 8.12~8.04(m, 3H), 7.93~7.82(m, 8H), 7.66(m, 1H), 7.54~7.25(m, 10H) 350 δ = 8.88(s, 1H), 8.30~8.21(m, 5H), 8.09~8.04(m, 2H), 7.92~7.82(m, 5H), 7.66(m, 2H), 7.54~7.32(m, 9H) 354 δ = 8.30(m, 2H), 7.92~7.82(m, 3H), 7.66(m, 1H), 7.54~7.26(m, 12H), 7.11~7.02(m, 8H), 6.80(m, 1H), 6.69(m, 2H), 6.51(m, 2H) 358 δ = 8.55(m, 2H), 8.30(m, 2H), 8.21(m, 1H), 8.12~8.04(m, 4H), 7.94~7.90(m, 5H), 7.63~7.29(m, 18H) 361 δ = 8.30(m, 2H), 8.21(m, 1H), 8.04(m, 1H), 7.92~7.89(m, 3H), 7.66(m, 1H), 7.54~7.32(m, 8H), 6.92~6.89(m, 4H), 6.77~6.69(m, 4H) 362 δ = 8.55(d, 1H), 8.21(m, 2H), 8.12(m, 1H), 8.04(m, 1H), 7.92~7.82(m, 5H), 7.71(m, 5H), 7.54~7.50(m, 3H), 7.38~7.32(m, 2H), 7.21~7.16(m, 6H), 6.97(m, 2H), 6.69(m, 2H) 366 δ = 8.30~8.21(m, 6H), 8.04(d, 1H), 7.92~7.85(m, 5H), 7.70(m, 1H), 7.66~7.32(m, 18H) 370 δ = 8.30~8.21(m, 4H), 8.04(d, 1H), 7.92~7.85(m, 6H), 7.70~7.66(m, 3H), 7.54~7.32(m, 16H) 374 δ = 8.28~8.21(m, 6H), 8.04(d, 1H), 7.92~7.81(m, 6H), 7.66~7.32(m, 17H) 378 δ = 8.55(m, 1H), 8.30~8.21(m, 4H), 8.00~7.87(m, 9H), 7.66~7.25(m, 23H) 381 δ = 8.30(d, 2H), 8.21(d, 1H), 8.04(d, 1H), 7.92~7.89(m, 3H), 7.66(m, 1H), 7.54~7.32(m, 7H), 7.05~7.02(m, 4H), 6.89~6.88(m, 2H), 6.73(m, 2H), 6.59~6.55(m, 3H), 1.72(s, 6H) 382 δ = 8.30(m, 2H), 8.21(m, 1H), 8.04(d, 1H), 7.90~7.89(m, 3H), 7.66(m, 1H), 7.54~7.32(m, 7H), 6.92~6.88(m, 6H), 6.77(m, 2H), 6.59(m, 3H) 386 δ = 8.30~8.21(m, 7H), 8.10~8.06(m, 3H), 7.92~7.89(m, 3H), 7.81(d, 1H), 7.66~7.32(m, 14H) 390 δ = 8.55(m, 1H), 8.30(m, 2H), 8.21(d, 1H), 8.08~8.04(m, 2H), 7.92~7.81(m, 9H), 7.66(m, 1H), 7.55~7.32(m, 15H) 394 δ = 8.49(d, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.12~8.04(m, 3H), 7.92~7.89(m, 3H), 7.57~7.29(m, 20H) 398 δ = 8.30(m, 2H), 8.21(d, 1H), 8.04(d, 1H), 7.92~7.89(m, 3H), 7.66(m, 3H), 7.57~7.32(m, 20H) 401 δ = 8.30~8.21(m, 6H), 8.04(m, 1H),

7.90~7.79(m, 11H), 7.66(m, 1H), 7.54~7.41(m, 14H) 402 δ = 8.30~8.21(m, 8H), 8.04(m, 1H),
7.90~7.79(m, 7H), 7.66(d, 1H), 7.54~7.25(m, 16H) 406 δ = 8.30~8.21(m, 6H), 8.04(d, 1H),
7.92~7.89(m, 5H), 7.66~7.32(m, 15H) 410 δ = 8.30~8.23(m, 8H), 8.04(d, 1H), 7.95~7.85(m, 7H),
7.75(m, 1H), 7.66~7.64(m, 3H), 7.54~7.32(m, 11H) 414 δ = 8.55(d, 1H), 8.30~8.21(m, 8H),
8.04(d, 1H), 7.94~7.87(m, 7H), 7.77(d, 1H), 7.66~7.25(m, 18H) 418 δ = 8.55(m, 1H),
8.30~8.21(m, 5H), 8.12(d, 1H), 8.04(d, 1H), 7.94~7.90(m, 4H), 7.66~7.25(m, 16H) 421 δ =
8.30~8.23(m, 5H), 8.04(d, 1H), 7.92~7.90(m, 3H), 7.79(m, 4H), 7.70~7.66(m, 2H), 7.57~7.25(m,
23H) 422 δ = 8.45(d, 1H), 8.30(m, 2H), 8.23~8.21(m, 2H), 8.08~7.82(m, 10H), 7.66(d, 1H),
7.54~7.41(m, 11H) 426 δ = 8.30~8.21(m, 4H), 8.04(d, 1H), 7.89~7.79(m, 7H), 7.66~7.32(m, 15H)
430 δ = 8.30~8.21(m, 6H), 8.04(d, 1H), 7.85~7.79(m, 10H), 7.66(m, 2H), 7.54~7.32(m, 10H) 446
8.45(m, 1H), 8.30(d, 2H), 8.21(d, 1H), 7.85(m, 8H), 7.66(t, 1H), 7.54~7.50(m, 16H), 7.25(m, 4H)
450 8.55(d, 1H), 8.45(d, 1H), 8.30~8.21(m, 5H), 8.02~7.87(m, 5H), 7.77(d, 1H), 7.52~7.25(m,
18H) 454 8.45(m, 1H), 8.30~8.21(m, 5H), 8.06~7.87(m, 7H), 7.61~7.41(m, 14H), 7.28~7.25(m,
3H) 458 8.45(d, 1H), 8.30~8.21(m, 5H), 8.02~7.81(m, 8H), 7.66(m, 2H), 7.54~7.32(m, 24H) 462
9.15(s, 1H), 8.93(d, 1H), 8.45(d, 1H), 8.30(m, 2H), 8.21~7.98(m, 9H), 8.12~7.82(m, 9H),
7.71~7.66(m, 3H), 7.54~7.47(m, 6H) 466 8.93(d, 1H), 8.45(d, 1H), 8.30(m, 2H), 8.21(m, 1H),
8.12(d, 2H), 8.02~7.82(m, 8H), 7.66(s, 1H), 7.54~7.74(m, 6H), 7.25(s, 4H) 470 8.88(s, 1H),
8.45(d, 1H), 8.30() 474 8.88(s, 1H), 8.45(d, 1H), 8.30(m, 2H), 8.21(d, 1H), 8.09(s, 1H),
8.02~7.92(m, 2H), 7.84~7.82(m, 2H), 7.66(m, 2H), 7.54~7.47(m, 9H) 478 8.45(d, 1H), 8.30(m,
2H), 8.21(d, 1H), 7.98~7.75(m, 7H), 7.66(m, 1H), 7.57~7.28(m, 16H) 482 8.25(d, 1H),
8.30~8.28(m, 6H), 8.02~7.81(m, 6H), 7.72~7.66(m, 3H), 7.54~7.41(m, 13H) 486 8.88(d, 1H),
8.45(d, 1H), 8.30(m, 4H), 8.10~8.06(m, 4H), 7.98~7.92(m, 2H), 7.81(m, 1H), 7.69(m, 1H),
7.54~7.47(m, 8H), 7.35(d, 2H) 490 8.45(d, 1H), 8.30(m, 2H), 8.21(d, 1H), 8.12(m, 1H),
8.02~7.92(m, 3H), 7.63~7.50(m, 13H), 7.29(m, 2H) 494 8.45(d, 1H), 8.30(m, 2H), 8.21(m, 2H),
8.12(m, 2H), 7.98 498 9.09(s, 1H), 8.49~8.45(m, 2H), 8.30~8.21(m, 7H), 8.04~7.92(m, 8H),
7.73(m, 3H), 7.58~7.41(m, 15H) 502 8.55(d, 1H), 8.45(d, 1H), 8.30~8.21(m, 4H), 8.00~7.87(m,
8H), 7.77~7.45(m, 21H), 7.33~7.25(m, 2H) 506 8.45(d, 1H), 8.30~8.21(m, 6H), 8.02~7.92(m, 3H),
7.70~7.41(m, 26H) 510 8.45(d, 1H), 8.30~8.21(m, 8H), 8.02~7.92(m, 3H), 7.54~7.41(m, 16H),
7.25(s, 4H) 514 8.45(d, 2H), 8.30~8.21(m, 6H), 8.02~7.94(m, 5H), 7.82(m, 1H), 7.66~7.41(m,
16H) 518 8.45~8.41(m, 2H), 8.30~8.21(m, 6H), 8.00~7.86(m, 9H), 7.58~7.41(m, 18H) 522 8.55(d,
1H), 8.45(d, 1H), 8.37~8.28(m, 6H), 8.02~7.87(m, 5H), 7.77(s, 1H), 7.69~7.33(m, 18H) 526
8.45(d, 1H), 8.30~8.21(m, 6H), 8.02~7.92(m, 3H), 7.79~7.48(m, 20H), 7.25(s, 4H) 530 8.45(d,
1H), 8.30~8.21(m, 6H), 8.02~7.92(m, 3H), 7.75~7.41(m, 19H), 7.25(s, 4H) 534 8.55(d, 1H),
8.45(d, 1H), 8.30~8.21(m, 8H), 8.02~7.92(m, 5H), 7.77(s, 1H), 7.66~7.25(m, 20H) 538 8.45(d,
1H), 8.30(m, 2H), 8.23~8.21(m, 2H), 8.02~7.79(m, 7H), 7.66~7.32(m, 15H) 542 8.45(d, 1H),
8.30(m, 2H), 8.23~8.21(m, 2H), 8.02~7.92(m, 3H), 7.79(m, 4H), 7.66(m, 1H), 7.54~7.41(m, 12H)
546 8.45(d, 1H), 8.30~8.21(m, 6H), 8.02~7.89(m, 4H), 7.79~7.75(m, 3H), 7.66~7.38(m, 17H) 550
8.45(d, 1H), 8.30~8.21(m, 6H), 8.02~7.95(m, 5H), 7.79~7.75(m, 3H), 7.66~7.64(m, 2H),
7.54~7.32(m, 13H) 554 8.45(d, 1H), 8.30~8.27(m, 5H), 8.12(d, 1H), 7.85~7.98(m, 5H),
7.52~7.41(m, 14H), 7.25(d, 2H) 558 8.45(d, 2H), 8.30(m, 4H), 8.23~8.21(m, 2H), 8.02~7.92(m,
5H), 7.79~7.82(m, 3H), 7.66(m, 1H), 7.54~7.41(m, 12H), 7.25(m, 2H) 562 8.45(s, 1H),
8.30~8.27(m, 5H), 8.12(m, 1H), 7.98~7.81(m, 6H), 7.66(m, 1H), 7.54~7.32(m, 12H) 566 8.45(d,
1H), 8.30~8.27(m, 5H), 8.12(m, 1H), 8.03~7.81(m, 8H), 7.66(m, 1H), 7.54~7.32(m, 14H) 570
8.45(d, 2H), 8.28~8.27(m, 5H), 8.12(d, 1H), 8.03~7.82(m, 8H), 7.56~7.41(m, 12H), 7.25(m, 2H)
574 8.55(m, 1H), 8.45(m, 1H), 8.30~8.27(m, 5H), 8.12(m, 2H), 8.03~7.92(m, 4H), 7.63(m, 1H),
7.54~7.25(m, 13H) 578 9.15(s, 1H), 8.93(m, 2H), 8.45(m, 1H), 8.30~8.27(m, 3H), 8.03~7.82(m,
12H), 7.54~7.41(m, 6H) 582 8.45(m, 2H), 8.30(m, 2H), 8.11~7.98(m, 8H), 7.54~7.47(m, 8H) 586
9.30(d, 1H), 9.15(s, 2H), 8.99(m, 1H), 8.81(s, 1H), 8.53(m, 1H), 8.45(m, 1H), 8.30~8.27(m, 3H),
8.12(m, 1H), 8.03~7.88(m, 4H), 7.70(m, 1H), 7.54~7.41(m, 11H), 7.14(m, 1H) 590 8.45(m, 1H),
8.30~8.27(m, 7H), 8.12(m, 1H), 8.03~7.81(m, 8H), 7.71~7.72(m, 2H), 7.54~7.41(m, 13H) 594

8.88(s, 1H), 8.45(d, 1H), 8.30~8.27(m, 5H), 8.12~7.98(m, 5H), 7.84~7.82(d, 2H), 7.66(s, 1H),
7.54~7.47(m, 9H) 598 8.55(m, 1H), 8.43(d, 1H), 8.30~8.27(m, 3H), 8.12(m, 2H), 8.03~7.92(m,
4H), 7.79(m, 1H), 7.68~7.25(m, 11H), 7.52~7.29(m, 3H) 602 8.45(m, 1H), 8.30~8.27(m, 3H),
8.12(m, 1H), 8.03~7.98(m, 3H), 7.54~7.47(m, 8H), 7.05~7.02(m, 4H), 6.79~6.69(m, 4H), 6.55(m,
2H), 1.72(s, 2H) 606 8.45(d, 1H), 8.27~8.30(m, 3H), 8.12(m, 1H), 7.98~7.75(m, 7H),
7.54~7.19(m, 16H) 610 8.72(m, 1H), 8.57(m, 1H), 8.45~8.41(m, 2H), 8.30~8.22(m, 6H),
8.10~7.92(m, 10H), 7.81(m, 1H), 7.73(m, 2H), 7.58~7.50(m, 11H), 7.35(m, 2H) 614 8.55(m, 1H),
8.45(m, 1H), 8.30~8.24(m, 4H), 8.12(m, 1H), 8.00~7.87(m, 8H), 7.77~7.69(m, 4H), 7.59~7.50(m,
16H), 7.33~7.25(m, 2H) 618 8.55(m, 1H), 8.45(d, 1H), 8.30~8.24(m, 6H), 8.12(m, 1H),
8.03~7.92(m, 5H), 7.77(m, 1H), 7.70~7.69(m, 2H), 7.58~7.25(m, 18H) 622 8.45(d, 1H),
8.30~8.24(m, 4H), 8.12(m, 1H), 7.85~7.98(m, 7H), 7.70(d, 1H), 7.57~7.41(m, 18H), 7.25(m, 4H)
626 8.45(d, 1H), 8.27~8.24(m, 4H), 8.12(d, 1H), 8.03~7.81(m, 6H), 7.66(m, 2H), 7.57~7.32(m,
16H) 630 8.55(d, 1H), 8.45~8.41(m, 2H), 8.30~8.28(m, 8H), 8.03~7.80(m, 10H), 7.55~7.41(m,
15H) 634 8.45(m, 1H), 8.28~8.24(m, 6H), 8.12(d, 1H), 8.03~7.81(m, 6H), 7.70~7.66(m, 2H),
7.57~7.32(m, 15H) 638 8.49~8.45(m, 2H), 8.30~8.27(m, 3H), 8.12~7.92(m, 6H), 7.70(m, 1H),
7.58~7.45(m, 17H), 7.29(m, 1H) 642 8.45(d, 1H), 8.30~8.27(m, 3H), 8.12(m, 1H), 8.03~7.92(m,
3H), 7.54~7.44(m, 7H), 6.92~6.88(m, 6H), 6.77(m, 2H), 6.59(m, 3H) 646 8.45(d, 1H),
8.30~8.26(m, 7H), 8.10~7.92(m, 7H), 7.81(m, 1H), 7.54~7.47(m, 11H) 650 8.92(m, 2H), 8.45(d,
1H), 8.30~8.27(m, 3H), 8.12(m, 3H), 8.03~7.82(m, 8H), 7.70(d, 1H), 7.57~7.48(m, 9H) 654
8.45(d, 2H), 8.28~8.23(m, 6H), 8.12(d, 1H), 8.03~7.92(m, 4H), 7.82(m, 1H), 7.52~7.41(m, 12H)
658 8.45(m, 2H), 8.30~8.23(m, 8H), 8.12(m, 1H), 7.98~7.85(m, 8H), 7.56~7.41(m, 12H) 662
8.45(m, 1H), 8.30~8.23(m, 8H), 8.12(d, 1H), 8.03~7.85(m, 5H), 7.54~7.41(m, 14H) 666 8.45(d,
1H), 8.30~8.23(m, 8H), 8.12(m, 1H), 8.03~7.85(m, 7H), 7.54~7.47(m, 12H), 7.25(s, 4H) 670
8.45(d, 1H), 8.30~8.23(m, 6H), 8.12(d, 1H), 8.03~7.92(m, 5H), 7.75(d, 1H), 7.66~7.64(m, 2H),
7.54~7.38(m, 11H) 674 8.45(d, 1H), 8.30~8.23(m, 4H), 8.12(d, 1H), 8.03~7.79(m, 7H),
7.54~7.32(m, 15H) 678 8.55(d, 1H), 8.45(d, 1H), 8.30~8.23(m, 4H), 8.12(d, 1H), 8.03~7.87(m,
5H), 7.79~7.77(m, 3H), 7.69(d, 1H), 7.58~7.41(m, 16H) 682 8.45(d, 1H), 8.30~8.23(m, 8H),
8.12(d, 1H), 7.98~7.79(m, 7H), 7.54~7.41(m, 12H) 686 8.45(d, 1H), 8.30~8.23(m, 4H), 8.12(d,
1H), 8.03~7.89(m, 4H), 7.79~7.75(m, 3H), 7.66~7.38(m, 15H) 690 8.45(d, 1H), 8.30~8.23(m, 6H),
8.12(d, 1H), 7.92~7.79(m, 10H), 7.66(d, 1H), 7.54~7.32(m, 12H) 694 8.45(d, 1H), 8.30~8.23(m,
9H), 8.12(d, 1H), 8.03~7.85(m, 5H), 7.75~7.70(m, 2H), 7.54~7.41(m, 16H) 698 8.45(d, 1H),
8.30~8.23(m, 6H), 8.12(d, 1H), 8.06~7.79(m, 8H), 7.61~7.41(m, 13H), 7.28(m, 1H), 1.72(m, 6H)
702 8.45(d, 1H), 8.30~8.21(m, 7H), 8.04~7.85(m, 6H), 7.54~7.41(m, 12H), 7.25(m, 2H) 706
8.45(d, 1H), 8.30~8.21(m, 5H), 8.04~7.58(m, 7H), 7.66(d, 1H), 7.54~7.32(m, 15H) 710 8.45(d,
1H), 8.30~8.21(m, 5H), 8.04~7.81(m, 9H), 7.66(m, 1H), 7.54~7.32(m, 15H) 714 8.55(d, 1H),
8.45(m, 1H), 8.30~8.21(m, 5H), 8.04~7.85(m, 8H), 7.77(d, 1H), 7.69(d, 1H), 7.58~7.25(m, 18) 718
8.55(d, 1H), 8.45(d, 1H), 8.30~8.21(m, 5H), 8.12(m, 1H), 8.04~7.92(m, 5H), 7.63(m, 1H),
7.54~7.25(m, 13H) 722 8.45(d, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.044~7.92(m, 4H), 7.66(s, 3H),
7.54~7.41(m, 16H), 7.25(d, 4H) 726 8.83(m, 2H), 8.45~8.38(m, 4H), 8.21(m, 1H), 8.04~7.90(m,
4H), 7.58~7.47(m, 9H) 730 8.88(s, 1H), 8.45(d, 1H), 8.30(m, 4H), 8.21(m, 1H), 7.90~7.82(m, 7H),
7.66(s, 1H), 7.54~7.47(m, 9H) 734 8.49~8.45(m, 2H), 8.30(m, 2H), 8.21(m, 1H), 8.12~7.90(m,
6H), 7.63~7.45(m, 14H), 7.29~7.25(m, 5H) 738 8.45(m, 2H), 8.30(m, 2H), 7.98~7.92(m, 2H),
7.54~7.47(m, 6H), 7.33~7.26(m, 6H), 7.11~6.98(m, 6H), 6.80(s, 1H), 6.69(m, 2H), 6.51(m, 2H)
742 9.09(s, 2H), 8.49~8.45(m, 3H), 8.30(m, 2H), 8.21(m, 1H), 8.04~7.85(m, 12H), 7.59~7.54(m,
10H), 7.25(m, 2H) 746 8.45(d, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.04~7.85(m, 7H), 7.75(m, 1H),
7.66~7.25(m, 18H) 750 9.09(s, 1H), 8.49~8.45(m, 3H), 8.30(m, 2H), 8.21(m, 1H), 8.04~7.94(m,
9H), 7.82(m, 1H), 7.59~7.47(m, 11H) 754 8.45(d, 1H), 8.30~8.21(m, 6H), 8.04~7.81(m, 7H),
7.70~7.66(m, 2H), 7.57~7.32(m, 14H) 758 8.55(m, 1H), 8.45(m, 1H), 8.30~8.21(m, 6H),
8.04~7.87(m, 6H), 7.77(s, 1H), 7.70~7.69(m, 2H), 7.58~7.41(m, 18H) 762 8.45(m, 1H), 8.30(m,
2H), 8.21(m, 1H), 8.04~7.92(m, 4H), 7.54~7.44(m, 7H), 7.05~7.02(m, 4H), 6.89~6.88(m, 2H),

6.73(m, 2H), 6.55(m, 2H) 766 8.45(d, 1H), 8.30~8.28(m, 6H), 7.79~7.75(m, 8H), 7.54~7.41(m, 15H) 770 8.49~8.45(m, 2H), 8.30(m, 2H), 8.21(m, 1H), 8.12~7.90(m, 6H), 7.70(s, 1H), 7.58~7.45(m, 17H), 7.29(m, 1H) 774 8.45~8.41(m, 3H), 8.30(m, 2H), 8.21~8.20(m, 2H), 8.04~7.98(m, 4H), 7.70(m, 1H), 7.54~7.41(m, 12H) 778 8.45(d, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.12(m, 2H), 8.04~7.90(m, 4H), 7.70(s, 1H), 7.63~7.47(m, 15H), 7.29(m, 2H) 782 8.45(d, 1H), 8.30~8.21(m, 5H), 8.04~7.92(m, 4H), 7.75~7.70(m, 3H), 7.57~7.41(m, 20H) 786 8.41~8.54(m, 2H), 8.30~8.21(m, 6H), 8.04~7.75(m, 11H), 7.57~7.41(m, 15H) 790 8.45(d, 1H), 8.30~8.21(m, 7H), 8.10~7.90(m, 7H), 7.81(d, 1H), 7.60~7.54(m, 11H) 794 8.45(d, 1H), 8.30~8.21(m, 8H), 8.11~7.98(m, 10H), 7.54~7.41(m, 11H) 898 8.55(m, 1H), 8.45(m, 1H), 8.30~8.21(m, 7H), 8.12(m, 1H), 8.04~7.92(m, 7H), 7.63(m, 1H), 7.54~7.25(m, 14H) 802 8.55(m, 1H), 8.45(d, 1H), 8.30~8.21(m, 6H), 8.04~7.94(m, 6H), 7.77(s, 1H), 7.69(d, 1H), 7.58~7.25(m, 16H) 806 8.45(d, 1H), 8.30(m, 2H), 8.23~8.21(m, 2H), 8.04~7.89(m, 6H), 7.79~7.75(m, 3H), 7.66~7.64(m, 2H), 7.54~7.38(m, 10H) 810 8.55(d, 1H), 8.45(m, 1H), 8.30~8.21(m, 5H), 8.12(m, 1H), 8.04~7.92(m, 5H), 7.63(m, 1H), 7.54~7.25(m, 14H) 814 8.45(d, 1H), 8.30~8.23(m, 7H), 8.04~7.85(m, 6H), 7.75~7.70(m, 3H), 7.57~7.41(m, 20H) 818 8.45(d, 1H), 8.30(m, 2H), 8.23~8.21(m, 2H), 8.04~7.94(m, 6H), 7.82~7.79(m, 3H), 7.54~7.41(m, 12H) 822 8.55(m, 1H), 8.45(d, 1H), 8.30(m, 2H), 8.21(m, 1H), 8.12(m, 1H), 8.04~7.90(m, 5H), 7.79(m, 2H), 7.63(m, 1H), 7.54~7.25(m, 14H) 826 8.45(m, 1H), 8.300~8.21(m, 6H), 8.04~7.79(m, 11H), 7.66(m, 1H), 7.54~7.32(m, 12H) 830 8.78(m, 2H), 8.50~8.45(m, 3H), 8.28~8.23(m, 6H), 7.85~7.79(m, 8H), 7.51~7.41(m, 10H)

EXPERIMENTAL EXAMPLE

Experimental Example 1

(147) 1) Manufacture of Organic Light Emitting Device

(148) A glass substrate on which indium tin oxide (ITO) was coated as a thin film to a thickness of 1,500 Å was cleaned with distilled water ultrasonic waves. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents such as acetone, methanol and isopropyl alcohol, then dried, and UVO treatment was conducted for 5 minutes using UV in a UV cleaner. After that, the substrate was transferred to a plasma cleaner (PT), and after conducting plasma treatment under vacuum for ITO work function and residual film removal, the substrate was transferred to a thermal deposition apparatus for organic deposition. On the transparent ITO electrode (anode), organic materials were formed in a single stack structure. The following HAT-CN was deposited to a thickness of 50 Å as a hole injection layer, and, as a hole transfer layer, DNTPD was doped to NPD by 10% or less and then deposited and formed to a thickness of 1500 Å, and TCTA was continuously deposited to a thickness of 200 Å. Subsequently, a light emitting layer including a t-Bu-perylene dopant in an ADN host was formed to a thickness of 250 Å. Then, Alq.sub.3, an electron transfer layer, was formed as a film to a thickness of 250 Å, and as an N-type charge generation layer, a compound described in the following Table 9 was doped with lithium, an alkali metal, to be formed as a film to a thickness of 100 Å. Then, Al, a cathode, was formed as a film to a thickness of approximately 1,000 Å, and as a result, an organic light emitting device was manufactured.

(149) ##STR00609## ##STR00610## ##STR00611## ##STR00612##

2) Driving Voltage and Light Emission Efficiency of Organic Light Emitting Device

(150) For each of the organic light emitting devices manufactured as above, electroluminescent (EL) properties were measured using M7000 manufactured by McScience Inc., and with the measurement results, T.sub.95 was measured when standard luminance was 750 cd/m.sup.2 through a lifetime measurement system (M6000) manufactured by McScience Inc. Results of measuring driving voltage, light emission efficiency, external quantum efficiency and color coordinate (CIE) of the white organic electroluminescent devices manufactured according to the present disclosure are as shown in Table 9.

(151) TABLE-US-00009 TABLE 9 Light Driving Emission Life- Com- Voltage Efficiency time pound (V) (cd/A) CIE (x, y) (T.sub.95) Example 1-1 1 4.53 7.21 (0.134, 0.104) 43 Example 1-2 4

4.18 7.13 (0.134, 0.104) 45 Example 1-3 8 4.27 7.30 (0.134, 0.105) 39 Example 1-4 12 4.21 6.98 (0.134, 0.103) 41 Example 1-5 15 4.97 6.56 (0.134, 0.101) 37 Example 1-6 18 4.57 7.00 (0.134, 0.101) 44 Example 1-7 21 5.12 6.48 (0.133, 0.102) 34 Example 1-8 25 4.32 7.11 (0.134, 0.100) 42 Example 1-9 28 5.11 6.42 (0.133, 0.101) 30 Example 1-10 32 5.27 6.55 (0.134, 0.101) 28 Example 1-11 35 5.68 6.42 (0.134, 0.101) 32 Example 1-12 38 5.88 6.30 (0.134, 0.101) 30 Example 1-13 41 5.14 6.51 (0.133, 0.101) 28 Example 1-14 44 5.89 6.18 (0.133, 0.102) 80 Example 1-15 47 5.90 6.11 (0.134, 0.100) 78 Example 1-16 50 5.28 6.78 (0.133, 0.101) 27 Example 1-17 53 5.44 6.37 (0.134, 0.101) 26 Example 1-18 58 5.68 6.32 (0.133, 0.100) 37 Example 1-19 61 5.40 6.48 (0.134, 0.100) 25 Example 1-20 63 5.38 6.50 (0.133, 0.101) 28 Example 1-21 67 5.33 6.48 (0.134, 0.101) 28 Example 1-22 73 5.36 6.54 (0.134, 0.099) 27 Example 1-23 76 5.34 6.55 (0.134, 0.101) 30 Example 1-24 78 5.68 6.43 (0.133, 0.101) 31 Example 1-25 81 5.44 6.40 (0.134, 0.102) 33 Example 1-26 83 5.67 6.39 (0.134, 0.101) 35 Example 1-27 86 4.87 5.99 (0.132, 0.101) 29 Example 1-28 90 5.68 6.47 (0.134, 0.103) 31 Example 1-29 92 5.68 6.50 (0.134, 0.101) 34 Example 1-30 95 5.66 6.77 (0.133, 0.102) 29 Example 1-31 101 5.73 6.41 (0.133, 0.101) 54 Example 1-32 103 5.64 6.40 (0.134, 0.100) 32 Example 1-33 107 5.71 6.52 (0.134, 0.101) 33 Example 1-34 110 5.68 6.55 (0.133, 0.100) 30 Example 1-35 113 5.70 6.40 (0.133, 0.101) 30 Example 1-36 115 5.23 6.33 (0.133, 0.101) 31 Example 1-37 122 5.78 6.44 (0.133, 0.101) 37 Example 1-38 124 5.73 6.62 (0.133, 0.099) 31 Example 1-39 126 5.22 6.38 (0.134, 0.101) 67 Example 1-40 130 5.11 6.52 (0.133, 0.101) 30 Example 1-41 135 5.27 6.55 (0.134, 0.101) 28 Example 1-42 138 5.68 6.40 (0.134, 0.101) 36 Example 1-43 142 5.88 6.34 (0.134, 0.101) 37 Example 1-44 144 4.53 7.21 (0.134, 0.104) 43 Example 1-45 147 4.18 7.13 (0.134, 0.104) 45 Example 1-46 149 4.27 7.30 (0.134, 0.105) 39 Example 1-47 152 4.21 6.98 (0.134, 0.103) 41 Example 1-48 155 4.97 6.56 (0.134, 0.101) 37 Example 1-49 161 4.57 7.00 (0.134, 0.101) 44 Example 1-50 164 5.12 6.48 (0.133, 0.102) 34 Example 1-51 168 4.32 7.11 (0.134, 0.100) 42 Example 1-52 172 5.11 6.42 (0.133, 0.101) 30 Example 1-53 174 5.27 6.55 (0.134, 0.101) 28 Example 1-54 178 5.68 6.42 (0.134, 0.101) 32 Example 1-55 181 5.88 6.30 (0.134, 0.101) 30 Example 1-56 186 5.14 6.51 (0.133, 0.101) 28 Example 1-57 190 5.89 6.18 (0.133, 0.102) 80 Example 1-58 194 5.90 6.11 (0.134, 0.100) 78 Example 1-59 198 5.28 6.78 (0.133, 0.101) 27 Example 1-60 201 5.44 6.37 (0.134, 0.101) 26 Example 1-61 202 5.68 6.32 (0.133, 0.100) 37 Example 1-62 206 5.40 6.48 (0.134, 0.100) 25 Example 1-63 210 5.38 6.50 (0.133, 0.101) 28 Example 1-64 214 5.33 6.48 (0.134, 0.101) 28 Example 1-65 218 5.36 6.54 (0.134, 0.099) 27 Example 1-66 221 5.34 6.55 (0.134, 0.101) 30 Example 1-67 222 5.68 6.43 (0.133, 0.101) 31 Example 1-68 226 5.44 6.40 (0.134, 0.102) 33 Example 1-69 230 5.67 6.39 (0.134, 0.101) 35 Example 1-70 234 4.87 5.99 (0.132, 0.101) 29 Example 1-71 238 5.68 6.47 (0.134, 0.103) 31 Example 1-72 241 5.68 6.50 (0.134, 0.101) 34 Example 1-73 242 5.66 6.77 (0.133, 0.102) 29 Example 1-74 246 5.64 6.40 (0.134, 0.100) 32 Example 1-75 250 5.71 6.52 (0.134, 0.101) 33 Example 1-76 254 5.68 6.55 (0.133, 0.100) 30 Example 1-77 258 5.70 6.40 (0.133, 0.101) 30 Example 1-78 261 5.23 6.33 (0.133, 0.101) 31 Example 1-79 262 5.78 6.44 (0.133, 0.101) 37 Example 1-80 266 5.73 6.62 (0.133, 0.099) 31 Example 1-81 270 5.22 6.38 (0.134, 0.101) 67 Example 1-82 274 5.11 6.52 (0.133, 0.101) 30 Example 1-83 278 5.27 6.55 (0.134, 0.101) 28 Example 1-84 280 5.68 6.40 (0.134, 0.101) 36 Example 1-85 282 5.88 6.34 (0.134, 0.101) 37 Example 1-86 286 4.53 7.21 (0.134, 0.104) 43 Example 1-87 290 4.18 7.13 (0.134, 0.104) 45 Example 1-88 294 4.27 7.30 (0.134, 0.105) 39 Example 1-89 298 5.68 6.43 (0.133, 0.101) 31 Example 1-90 301 5.44 6.40 (0.134, 0.102) 33 Example 1-91 302 5.67 6.39 (0.134, 0.101) 35 Example 1-92 306 4.87 5.99 (0.132, 0.101) 29 Example 1-93 310 5.68 6.47 (0.134, 0.103) 31 Example 1-94 314 5.68 6.50 (0.134, 0.101) 34 Example 1-95 318 5.66 6.77 (0.133, 0.102) 29 Example 1-96 321 5.64 6.40 (0.134, 0.100) 32 Example 1-97 322 5.71 6.52 (0.134, 0.101) 33 Example 1-98 326 5.68 6.55 (0.133, 0.100) 30 Example 1-99 330 5.70 6.40 (0.133, 0.101) 30 Example 1-100 334 5.23 6.33 (0.133, 0.101) 31 Example 1-101 338 5.78 6.44 (0.133, 0.101) 37 Example 1-102 341 5.73 6.62 (0.133, 0.099) 31 Example 1-103 342 5.22 6.38 (0.134, 0.101) 67

Example 1-104 346 5.73 6.62 (0.133, 0.099) 31 Example 1-105 350 5.22 6.38 (0.134, 0.101) 67
Example 1-106 354 5.11 6.52 (0.133, 0.101) 30 Example 1-107 358 5.27 6.55 (0.134, 0.101) 28
Example 1-108 361 5.68 6.40 (0.134, 0.101) 36 Example 1-109 362 5.88 6.34 (0.134, 0.101) 37
Example 1-110 366 4.53 7.21 (0.134, 0.104) 43 Example 1-111 370 4.18 7.13 (0.134, 0.104) 45
Example 1-112 374 4.27 7.30 (0.134, 0.105) 39 Example 1-113 378 4.21 6.98 (0.134, 0.103) 41
Example 1-114 381 4.97 6.56 (0.134, 0.101) 37 Example 1-115 382 4.57 7.00 (0.134, 0.101) 44
Example 1-116 386 5.12 6.48 (0.133, 0.102) 34 Example 1-117 390 4.32 7.11 (0.134, 0.100) 42
Example 1-118 394 5.68 6.42 (0.134, 0.101) 32 Example 1-119 398 5.88 6.30 (0.134, 0.101) 30
Example 1-120 401 5.14 6.51 (0.133, 0.101) 28 Example 1-121 402 5.89 6.18 (0.133, 0.102) 80
Example 1-122 406 5.90 6.11 (0.134, 0.100) 78 Example 1-123 410 5.28 6.78 (0.133, 0.101) 27
Example 1-124 414 5.44 6.37 (0.134, 0.101) 26 Example 1-125 418 5.68 6.32 (0.133, 0.100) 37
Example 1-126 421 5.40 6.48 (0.134, 0.100) 25 Example 1-127 422 5.38 6.50 (0.133, 0.101) 28
Example 1-128 426 5.33 6.48 (0.134, 0.101) 28 Example 1-129 430 5.36 6.54 (0.134, 0.099) 27
Example 1-130 446 5.34 6.55 (0.134, 0.101) 30 Example 1-131 450 5.68 6.43 (0.133, 0.101) 31
Example 1-132 454 5.44 6.40 (0.134, 0.102) 33 Example 1-133 458 5.67 6.39 (0.134, 0.101) 35
Example 1-134 462 4.87 5.99 (0.132, 0.101) 29 Example 1-135 466 5.68 6.47 (0.134, 0.103) 31
Example 1-136 470 5.68 6.50 (0.134, 0.101) 34 Example 1-137 474 5.66 6.77 (0.133, 0.102) 29
Example 1-138 478 5.73 6.41 (0.133, 0.101) 54 Example 1-139 482 5.64 6.40 (0.134, 0.100) 32
Example 1-140 486 5.71 6.52 (0.134, 0.101) 33 Example 1-141 490 5.68 6.55 (0.133, 0.100) 30
Example 1-142 494 5.70 6.40 (0.133, 0.101) 30 Example 1-143 498 5.23 6.33 (0.133, 0.101) 31
Example 1-144 502 5.78 6.44 (0.133, 0.101) 37 Example 1-145 506 5.73 6.62 (0.133, 0.099) 31
Example 1-146 510 5.22 6.38 (0.134, 0.101) 67 Example 1-147 514 5.11 6.52 (0.133, 0.101) 30
Example 1-148 518 5.27 6.55 (0.134, 0.101) 28 Example 1-149 522 5.68 6.40 (0.134, 0.101) 36
Example 1-150 526 5.88 6.34 (0.134, 0.101) 37 Example 1-151 530 4.53 7.21 (0.134, 0.104) 43
Example 1-152 534 4.18 7.13 (0.134, 0.104) 45 Example 1-153 538 4.27 7.30 (0.134, 0.105) 39
Example 1-154 542 4.21 6.98 (0.134, 0.103) 41 Example 1-155 546 4.97 6.56 (0.134, 0.101) 37
Example 1-156 550 4.57 7.00 (0.134, 0.101) 44 Example 1-157 554 5.12 6.48 (0.133, 0.102) 34
Example 1-158 558 4.32 7.11 (0.134, 0.100) 42 Example 1-159 562 5.11 6.42 (0.133, 0.101) 30
Example 1-160 566 5.27 6.55 (0.134, 0.101) 28 Example 1-161 570 5.68 6.42 (0.134, 0.101) 32
Example 1-162 574 5.73 6.62 (0.133, 0.099) 31 Example 1-163 578 5.22 6.38 (0.134, 0.101) 67
Example 1-164 582 5.73 6.62 (0.133, 0.099) 31 Example 1-165 586 5.22 6.38 (0.134, 0.101) 67
Example 1-166 590 5.11 6.52 (0.133, 0.101) 30 Example 1-167 594 5.27 6.55 (0.134, 0.101) 28
Example 1-168 598 5.68 6.40 (0.134, 0.101) 36 Example 1-169 602 5.88 6.34 (0.134, 0.101) 37
Example 1-170 606 4.53 7.21 (0.134, 0.104) 43 Example 1-171 610 5.89 6.18 (0.133, 0.102) 80
Example 1-172 614 5.90 6.11 (0.134, 0.100) 78 Example 1-173 618 5.28 6.78 (0.133, 0.101) 27
Example 1-174 622 5.44 6.37 (0.134, 0.101) 26 Example 1-175 626 5.68 6.32 (0.133, 0.100) 37
Example 1-176 630 5.40 6.48 (0.134, 0.100) 25 Example 1-177 634 5.38 6.50 (0.133, 0.101) 28
Example 1-178 638 5.33 6.48 (0.134, 0.101) 28 Example 1-179 642 5.36 6.54 (0.134, 0.099) 27
Example 1-180 646 5.34 6.55 (0.134, 0.101) 30 Example 1-181 650 5.68 6.43 (0.133, 0.101) 31
Example 1-182 654 5.44 6.40 (0.134, 0.102) 33 Example 1-183 658 5.67 6.39 (0.134, 0.101) 35
Example 1-184 662 4.87 5.99 (0.132, 0.101) 29 Example 1-185 666 5.68 6.47 (0.134, 0.103) 31
Example 1-186 670 5.68 6.50 (0.134, 0.101) 34 Example 1-187 674 5.66 6.77 (0.133, 0.102) 29
Example 1-188 678 5.73 6.41 (0.133, 0.101) 54 Example 1-189 682 5.64 6.40 (0.134, 0.100) 32
Example 1-190 686 5.71 6.52 (0.134, 0.101) 33 Example 1-191 690 5.68 6.55 (0.133, 0.100) 30
Example 1-192 694 5.70 6.40 (0.133, 0.101) 30 Example 1-193 698 5.23 6.33 (0.133, 0.101) 31
Example 1-194 702 5.78 6.44 (0.133, 0.101) 37 Example 1-195 706 5.73 6.62 (0.133, 0.099) 31
Example 1-196 710 5.22 6.38 (0.134, 0.101) 67 Example 1-197 714 5.11 6.52 (0.133, 0.101) 30
Example 1-198 718 5.27 6.55 (0.134, 0.101) 28 Example 1-199 722 5.68 6.40 (0.134, 0.101) 36
Example 1-200 726 5.88 6.34 (0.134, 0.101) 37 Example 1-201 730 4.53 7.21 (0.134, 0.104) 43
Example 1-202 734 4.18 7.13 (0.134, 0.104) 45 Example 1-203 738 4.27 7.30 (0.134, 0.105) 39

Example 1-204 742 4.21 6.98 (0.134, 0.103) 41 Example 1-205 746 4.97 6.56 (0.134, 0.101) 37
 Example 1-206 750 4.57 7.00 (0.134, 0.101) 44 Example 1-207 754 5.12 6.48 (0.133, 0.102) 34
 Example 1-208 758 4.32 7.11 (0.134, 0.100) 42 Example 1-209 762 5.11 6.42 (0.133, 0.101) 30
 Example 1-210 766 5.27 6.55 (0.134, 0.101) 28 Example 1-211 770 5.68 6.42 (0.134, 0.101) 32
 Example 1-212 774 4.57 7.00 (0.134, 0.101) 44 Example 1-213 778 5.68 6.47 (0.134, 0.103) 31
 Example 1-214 782 5.68 6.50 (0.134, 0.101) 34 Example 1-215 786 5.66 6.77 (0.133, 0.102) 29
 Example 1-216 790 5.73 6.41 (0.133, 0.101) 54 Example 1-217 794 5.64 6.40 (0.134, 0.100) 32
 Example 1-218 898 5.71 6.52 (0.134, 0.101) 33 Example 1-219 802 5.68 6.55 (0.133, 0.100) 30
 Example 1-220 806 5.70 6.40 (0.133, 0.101) 30 Example 1-221 810 5.23 6.33 (0.133, 0.101) 31
 Example 1-222 814 5.78 6.44 (0.133, 0.101) 37 Example 1-223 818 5.73 6.62 (0.133, 0.099) 31
 Example 1-224 822 5.70 6.40 (0.133, 0.101) 30 Example 1-225 826 5.88 6.34 (0.134, 0.101) 37
 Example 1-226 830 4.27 7.30 (0.134, 0.105) 39 Comparative A 5.82 2.23 (0.134, 0.110) 27
 Example 1-1 Comparative B 5.80 3.32 (0.134, 0.111) 29 Example 1-2 Comparative C 5.84 2.39
 (0.134, 0.111) 25 Example 1-3 Comparative D 8.60 1.23 (0.134, 0.100) 21 Example 1-4
 Comparative E 7.81 3.23 (0.128, 0.111) 10 Example 1-5 Comparative F 6.87 2.30 (0.125, 0.097) 12
 Example 1-6

(152) As seen from the results of Table 9, the organic light emitting device using the charge generation layer material of the white organic light emitting device of the present disclosure had lower driving voltage and improved light emission efficiency compared to Comparative Examples 1-1 to 1-6.

(153) Such a result is considered to be due to the fact that the compound of the present disclosure used as an N-type charge generation layer formed with the disclosed skeleton having proper length and strength, and flatness and a proper hetero-compound capable of binding to metals forms a gap state in the N-type charge generation layer by doping an alkali metal or an alkaline earth metal thereto, and electrons produced from a P-type charge generation layer are readily injected into an electron transfer layer through the gap state produced in the N-type charge generation layer. Accordingly, the P-type charge generation layer may favorably inject and transfer electrons to the N-type charge generation layer, and as a result, driving voltage was lowered, and efficiency and lifetime were improved in the organic light emitting device.

Experimental Example 2

(154) 1) Manufacture of Organic Light Emitting Device

(155) A glass substrate on which indium tin oxide (ITO) was coated as a thin film to a thickness of 1,500 Å was cleaned with distilled water ultrasonic waves. After the cleaning with distilled water was finished, the substrate was ultrasonic cleaned with solvents such as acetone, methanol and isopropyl alcohol, then dried, and UVO treatment was conducted for 5 minutes using UV in a UV cleaner. After that, the substrate was transferred to a plasma cleaner (PT), and after conducting plasma treatment under vacuum for ITO work function and residual film removal, the substrate was transferred to a thermal deposition apparatus for organic deposition.

(156) On the transparent ITO electrode (anode), organic materials were formed in a 2-stack white organic light emitting device (WOLED) structure. As for the first stack, TAPC was thermal vacuum deposited first to a thickness of 300 Å to form a hole transfer layer. After forming the hole transfer layer, a light emitting layer was thermal vacuum deposited thereon as follows. As the light emitting layer, TCz1, a host, was 8% doped with FIrpic, a blue phosphorescent dopant, and deposited to 300 Å. After forming an electron transfer layer to 400 Å using TmPyPB, a compound described in the following Table 10 was 20% doped with Cs.sub.2CO.sub.3 and formed to 100 Å as a charge generation layer.

(157) As for the second stack, MoO.sub.3 was thermal vacuum deposited first to a thickness of 50 Å to form a hole injection layer. A hole transfer layer, a common layer, was formed to 100 Å by 20% doping MoO.sub.3 to TAPC, and then depositing the TAPC to 300 Å. A light emitting layer was formed thereon by 8% doping Ir(ppy).sub.3, a green phosphorescent dopant, to TCz1, a host,

and depositing the result to 300 Å, and then an electron transfer layer was formed to 600 Å using TmPyPB. Lastly, an electron injection layer was formed on the electron transfer layer by depositing lithium fluoride (LiF) to a thickness of 10 Å, and then a cathode was formed on the electron injection layer by depositing an aluminum (Al) cathode to a thickness of 1,200 Å, and as a result, an organic light emitting device was manufactured.

(158) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum sublimation purified under 10.sup.-8 torr to 10.sup.-6 torr for each material to be used in the OLED manufacture.

(159) ##STR00613## ##STR00614## ##STR00615##

2) Driving Voltage and Light Emission Efficiency of Organic Light Emitting Device

(160) For each of the organic light emitting devices manufactured as above, electroluminescent (EL) properties were measured using M7000 manufactured by McScience Inc., and with the measurement results, T.sub.95 was measured when standard luminance was 3,500 cd/m.sup.2 through a lifetime measurement system (M6000) manufactured by McScience Inc. Results of measuring driving voltage, light emission efficiency, external quantum efficiency and color coordinate (CIE) of the white organic light emitting devices manufactured according to the present disclosure are as shown in the following Table 10.

(161) TABLE-US-00010 TABLE 10 Light Driving Emission Life- Com- Voltage Efficiency time
pound (V) (cd/A) CIE (x, y) (T.sub.95) Example 2-1 1 7.23 68.95 (0.218, 0.427) 35 Example 2-2 4
7.02 69.45 (0.220, 0.431) 49 Example 2-3 8 7.07 64.88 (0.200, 0.421) 32 Example 2-4 12 7.11
67.23 (0.205, 0.411) 40 Example 2-5 15 7.65 63.21 (0.221, 0.434) 33 Example 2-6 18 7.01 69.82
(0.220, 0.440) 43 Example 2-7 21 7.67 58.98 (0.219, 0.411) 35 Example 2-8 25 7.10 69.45 (0.219,
0.429) 40 Example 2-9 28 7.66 59.22 (0.215, 0.411) 30 Example 2-10 32 7.77 60.94 (0.211, 0.419)
32 Example 2-11 35 8.24 58.26 (0.209, 0.419) 31 Example 2-12 38 8.01 59.11 (0.207, 0.409) 33
Example 2-13 41 7.40 63.66 (0,208, 0.415) 29 Example 2-14 44 8.06 58.03 (0.208, 0.412) 52
Example 2-15 47 8.24 57.98 (0.208, 0.411) 56 Example 2-16 50 7.96 60.77 (0.208, 0.412) 27
Example 2-17 53 7.88 61.29 (0.209, 0.412) 30 Example 2-18 58 8.13 59.01 (0.207, 0.411) 49
Example 2-19 61 8.00 59.13 (0.231, 0.440) 34 Example 2-20 63 7.82 63.22 (0.232, 0.422) 26
Example 2-21 67 7.99 63.88 (0.230, 0.420) 30 Example 2-22 73 8.25 61.12 (0.223, 0.433) 36
Example 2-23 76 7.76 63.47 (0.222, 0.435) 31 Example 2-24 78 8.22 60.84 (0.218, 0.421) 30
Example 2-25 81 8.37 59.91 (0.220, 0.421) 30 Example 2-26 83 8.25 62.56 (0.224, 0.429) 32
Example 2-27 86 7.55 60.99 (0.215, 0.422) 27 Example 2-28 90 8.33 61.11 (0,214, 0.420) 33
Example 2-29 92 7.74 63.32 (0.230, 0.439) 37 Example 2-30 95 7.71 65.97 (0.208, 0.412) 29
Example 2-31 101 8.52 57.04 (0.231, 0.418) 70 Example 2-32 103 7.78 60.01 (0.208, 0.412) 32
Example 2-33 107 8.01 59.97 (0.209, 0.411) 41 Example 2-34 110 7.71 64.37 (0.208, 0.412) 30
Example 2-35 113 7.82 54.67 (0.233, 0.419) 30 Example 2-36 115 7.54 63.58 (0.208, 0.412) 31
Example 2-37 122 7.88 59.36 (0.207, 0.417) 37 Example 2-38 124 7.71 62.03 (0.220, 0.412) 29
Example 2-39 126 8.38 59.80 (0.231, 0.423) 71 Example 2-40 130 7.91 60.22 (0.215, 0.411) 28
Example 2-41 135 7.97 60.94 (0.211, 0.419) 31 Example 2-42 138 8.24 58.26 (0.209, 0.419) 35
Example 2-43 142 8.01 59.11 (0.207, 0.409) 37 Example 2-44 144 7.23 68.95 (0.218, 0.427) 35
Example 2-45 147 7.02 69.45 (0.220, 0.431) 49 Example 2-46 149 7.07 64.88 (0.200, 0.421) 32
Example 2-47 152 7.11 67.23 (0.205, 0.411) 40 Example 2-48 155 7.65 63.21 (0.221, 0.434) 33
Example 2-49 161 7.01 69.82 (0.220, 0.440) 43 Example 2-50 164 7.67 58.98 (0.219, 0.411) 35
Example 2-51 168 7.10 69.45 (0.219, 0.429) 40 Example 2-52 172 7.66 59.22 (0.215, 0.411) 30
Example 2-53 174 7.77 60.94 (0.211, 0.419) 32 Example 2-54 178 8.24 58.26 (0.209, 0.419) 31
Example 2-55 181 8.01 59.11 (0.207, 0.409) 33 Example 2-56 186 7.40 63.66 (0,208, 0.415) 29
Example 2-57 190 8.06 58.03 (0.208, 0.412) 52 Example 2-58 194 8.24 57.98 (0.208, 0.411) 56
Example 2-59 198 7.96 60.77 (0.208, 0.412) 27 Example 2-60 201 7.88 61.29 (0.209, 0.412) 30
Example 2-61 202 8.13 59.01 (0.207, 0.411) 49 Example 2-62 206 8.00 59.13 (0.231, 0.440) 34
Example 2-63 210 7.82 63.22 (0.232, 0.422) 26 Example 2-64 214 7.40 63.66 (0,208, 0.415) 29

Example 2-65 218 8.06 58.03 (0.208, 0.412) 52 Example 2-66 221 8.24 57.98 (0.208, 0.411) 56
Example 2-67 222 7.96 60.77 (0.208, 0.412) 27 Example 2-68 226 7.88 61.29 (0.209, 0.412) 30
Example 2-69 230 8.13 59.01 (0.207, 0.411) 49 Example 2-70 234 8.00 59.13 (0.231, 0.440) 34
Example 2-71 238 7.82 63.22 (0.232, 0.422) 26 Example 2-72 241 7.99 63.88 (0.230, 0.420) 30
Example 2-73 242 8.25 61.12 (0.223, 0.433) 36 Example 2-74 246 7.76 63.47 (0.222, 0.435) 31
Example 2-75 250 8.22 60.84 (0.218, 0.421) 30 Example 2-76 254 8.37 59.91 (0.220, 0.421) 30
Example 2-77 258 8.25 62.56 (0.224, 0.429) 32 Example 2-78 261 7.55 60.99 (0.215, 0.422) 27
Example 2-79 262 7.66 59.22 (0.215, 0.411) 30 Example 2-80 266 7.77 60.94 (0.211, 0.419) 32
Example 2-81 270 8.24 58.26 (0.209, 0.419) 31 Example 2-82 274 8.01 59.11 (0.207, 0.409) 33
Example 2-83 278 7.40 63.66 (0.208, 0.415) 29 Example 2-84 280 8.06 58.03 (0.208, 0.412) 52
Example 2-85 282 8.24 57.98 (0.208, 0.411) 56 Example 2-86 286 7.96 60.77 (0.208, 0.412) 27
Example 2-87 290 7.88 61.29 (0.209, 0.412) 30 Example 2-88 294 8.13 59.01 (0.207, 0.411) 49
Example 2-89 298 8.00 59.13 (0.231, 0.440) 34 Example 2-90 301 7.82 63.22 (0.232, 0.422) 26
Example 2-91 302 7.40 63.66 (0.208, 0.415) 29 Example 2-92 306 7.74 63.32 (0.230, 0.439) 37
Example 2-93 310 7.71 65.97 (0.208, 0.412) 29 Example 2-94 314 8.52 57.04 (0.231, 0.418) 70
Example 2-95 318 7.78 60.01 (0.208, 0.412) 32 Example 2-96 321 8.01 59.97 (0.209, 0.411) 41
Example 2-97 322 7.71 64.37 (0.208, 0.412) 30 Example 2-98 326 7.82 54.67 (0.233, 0.419) 30
Example 2-99 330 7.54 63.58 (0.208, 0.412) 31 Example 2-100 334 7.88 59.36 (0.207, 0.417) 37
Example 2-101 338 7.71 62.03 (0.220, 0.412) 29 Example 2-102 341 8.38 59.80 (0.231, 0.423) 71
Example 2-103 342 7.91 60.22 (0.215, 0.411) 28 Example 2-104 346 7.97 60.94 (0.211, 0.419) 31
Example 2-105 350 8.24 58.26 (0.209, 0.419) 35 Example 2-106 354 7.88 61.29 (0.209, 0.412) 30
Example 2-107 358 8.13 59.01 (0.207, 0.411) 49 Example 2-108 361 8.00 59.13 (0.231, 0.440) 34
Example 2-109 362 7.82 63.22 (0.232, 0.422) 26 Example 2-110 366 7.40 63.66 (0.208, 0.415) 29
Example 2-111 370 7.74 63.32 (0.230, 0.439) 37 Example 2-112 374 7.71 65.97 (0.208, 0.412) 29
Example 2-113 378 8.52 57.04 (0.231, 0.418) 70 Example 2-114 381 7.78 60.01 (0.208, 0.412) 32
Example 2-115 382 8.01 59.97 (0.209, 0.411) 41 Example 2-116 386 7.71 64.37 (0.208, 0.412) 30
Example 2-117 390 7.82 54.67 (0.233, 0.419) 30 Example 2-118 394 7.54 63.58 (0.208, 0.412) 31
Example 2-119 398 7.11 67.23 (0.205, 0.411) 40 Example 2-120 401 7.65 63.21 (0.221, 0.434) 33
Example 2-121 402 7.01 69.82 (0.220, 0.440) 43 Example 2-122 406 7.67 58.98 (0.219, 0.411) 35
Example 2-123 410 7.10 69.45 (0.219, 0.429) 40 Example 2-124 414 7.66 59.22 (0.215, 0.411) 30
Example 2-125 418 7.77 60.94 (0.211, 0.419) 32 Example 2-126 421 8.24 58.26 (0.209, 0.419) 31
Example 2-127 422 8.01 59.11 (0.207, 0.409) 33 Example 2-128 426 7.40 63.66 (0.208, 0.415) 29
Example 2-129 430 8.06 58.03 (0.208, 0.412) 52 Example 2-130 446 8.24 57.98 (0.208, 0.411) 56
Example 2-131 450 7.96 60.77 (0.208, 0.412) 27 Example 2-132 454 7.88 61.29 (0.209, 0.412) 30
Example 2-133 458 8.13 59.01 (0.207, 0.411) 49 Example 2-134 462 8.00 59.13 (0.231, 0.440) 34
Example 2-135 466 7.82 63.22 (0.232, 0.422) 26 Example 2-136 470 7.78 60.01 (0.208, 0.412) 32
Example 2-137 474 8.01 59.97 (0.209, 0.411) 41 Example 2-138 478 7.71 64.37 (0.208, 0.412) 30
Example 2-139 482 7.82 54.67 (0.233, 0.419) 30 Example 2-140 486 7.54 63.58 (0.208, 0.412) 31
Example 2-141 490 7.11 67.23 (0.205, 0.411) 40 Example 2-142 494 7.65 63.21 (0.221, 0.434) 33
Example 2-143 498 7.01 69.82 (0.220, 0.440) 43 Example 2-144 502 7.67 58.98 (0.219, 0.411) 35
Example 2-145 506 7.10 69.45 (0.219, 0.429) 40 Example 2-146 510 7.66 59.22 (0.215, 0.411) 30
Example 2-147 514 7.77 60.94 (0.211, 0.419) 32 Example 2-148 518 8.24 58.26 (0.209, 0.419) 31
Example 2-149 522 8.01 59.11 (0.207, 0.409) 33 Example 2-150 526 7.55 60.99 (0.215, 0.422) 27
Example 2-151 530 7.66 59.22 (0.215, 0.411) 30 Example 2-152 534 7.77 60.94 (0.211, 0.419) 32
Example 2-153 538 8.24 58.26 (0.209, 0.419) 31 Example 2-154 542 8.01 59.11 (0.207, 0.409) 33
Example 2-155 546 7.40 63.66 (0.208, 0.415) 29 Example 2-156 550 8.06 58.03 (0.208, 0.412) 52
Example 2-157 554 8.24 57.98 (0.208, 0.411) 56 Example 2-158 558 7.96 60.77 (0.208, 0.412) 27
Example 2-159 562 7.88 61.29 (0.209, 0.412) 30 Example 2-160 566 8.13 59.01 (0.207, 0.411) 49
Example 2-161 570 8.00 59.13 (0.231, 0.440) 34 Example 2-162 574 7.82 63.22 (0.232, 0.422) 26
Example 2-163 578 7.40 63.66 (0.208, 0.415) 29 Example 2-164 582 7.74 63.32 (0.230, 0.439) 37

Example 2-165 586 7.71 65.97 (0.208, 0.412) 29 Example 2-166 590 8.52 57.04 (0.231, 0.418) 70
 Example 2-167 594 7.78 60.01 (0.208, 0.412) 32 Example 2-168 598 8.01 59.97 (0.209, 0.411) 41
 Example 2-169 602 7.71 64.37 (0.208, 0.412) 30 Example 2-170 606 7.82 54.67 (0.233, 0.419) 30
 Example 2-171 610 7.54 63.58 (0.208, 0.412) 31 Example 2-172 614 7.88 59.36 (0.207, 0.417) 37
 Example 2-173 618 7.71 62.03 (0.220, 0.412) 29 Example 2-174 622 7.82 63.22 (0.232, 0.422) 26
 Example 2-175 626 7.40 63.66 (0.208, 0.415) 29 Example 2-176 630 7.74 63.32 (0.230, 0.439) 37
 Example 2-177 634 7.71 65.97 (0.208, 0.412) 29 Example 2-178 638 8.52 57.04 (0.231, 0.418) 70
 Example 2-179 642 7.78 60.01 (0.208, 0.412) 32 Example 2-180 646 8.01 59.97 (0.209, 0.411) 41
 Example 2-181 650 7.71 64.37 (0.208, 0.412) 30 Example 2-182 654 7.82 54.67 (0.233, 0.419) 30
 Example 2-183 658 7.54 63.58 (0.208, 0.412) 31 Example 2-184 662 7.88 59.36 (0.207, 0.417) 37
 Example 2-185 666 7.71 62.03 (0.220, 0.412) 29 Example 2-186 670 7.66 59.22 (0.215, 0.411) 30
 Example 2-187 674 7.77 60.94 (0.211, 0.419) 32 Example 2-188 678 8.24 58.26 (0.209, 0.419) 31
 Example 2-189 682 8.01 59.11 (0.207, 0.409) 33 Example 2-190 686 7.55 60.99 (0.215, 0.422) 27
 Example 2-191 690 7.66 59.22 (0.215, 0.411) 30 Example 2-192 694 7.77 60.94 (0.211, 0.419) 32
 Example 2-193 698 7.45 71.80 (0.210, 0.418) 35 Example 2-194 702 7.11 67.23 (0.205, 0.411) 40
 Example 2-195 706 7.65 63.21 (0.221, 0.434) 33 Example 2-196 710 7.01 69.82 (0.220, 0.440) 43
 Example 2-197 714 7.67 58.98 (0.219, 0.411) 35 Example 2-198 718 7.10 69.45 (0.219, 0.429) 40
 Example 2-199 722 7.66 59.22 (0.215, 0.411) 30 Example 2-200 726 7.77 60.94 (0.211, 0.419) 32
 Example 2-201 730 8.24 58.26 (0.209, 0.419) 31 Example 2-202 734 8.01 59.11 (0.207, 0.409) 33
 Example 2-203 738 7.40 63.66 (0.208, 0.415) 29 Example 2-204 742 8.06 58.03 (0.208, 0.412) 52
 Example 2-205 746 8.24 57.98 (0.208, 0.411) 56 Example 2-206 750 7.96 60.77 (0.208, 0.412) 27
 Example 2-207 754 7.88 61.29 (0.209, 0.412) 30 Example 2-208 758 8.13 59.01 (0.207, 0.411) 49
 Example 2-209 762 8.00 59.13 (0.231, 0.440) 34 Example 2-210 766 7.82 63.22 (0.232, 0.422) 26
 Example 2-211 770 7.66 59.22 (0.215, 0.411) 30 Example 2-212 774 7.77 60.94 (0.211, 0.419) 32
 Example 2-213 778 8.24 58.26 (0.209, 0.419) 31 Example 2-214 782 8.01 59.11 (0.207, 0.409) 33
 Example 2-215 786 7.40 63.66 (0.208, 0.415) 29 Example 2-216 790 8.06 58.03 (0.208, 0.412) 52
 Example 2-217 794 8.24 57.98 (0.208, 0.411) 56 Example 2-218 898 7.96 60.77 (0.208, 0.412) 27
 Example 2-219 802 7.88 61.29 (0.209, 0.412) 30 Example 2-220 806 8.13 59.01 (0.207, 0.411) 49
 Example 2-221 810 7.71 64.37 (0.208, 0.412) 30 Example 2-222 814 7.82 54.67 (0.233, 0.419) 30
 Example 2-223 818 7.54 63.58 (0.208, 0.412) 31 Example 2-224 822 7.88 59.36 (0.207, 0.417) 37
 Example 2-225 826 7.71 62.03 (0.220, 0.412) 29 Example 2-226 830 7.82 63.22 (0.232, 0.422) 26
 Comparative A 2.82 23.23 (0.134, 0.110) 21 Example 2-1 Comparative B 3.80 32.32 (0.134, 0.111) 19
 Example 2-2 Comparative C 5.64 15.65 (0.128, 0.099) 8 Example 2-3 Comparative D 3.20 18.22 (0.119, 0.100) 11
 Example 2-4 Comparative E 6.54 10.88 (0.128, 0.111) 9 Example 2-5 Comparative F 2.34 9.86 (0.125, 0.097) 15
 Example 2-6

(162) As seen from the results of Table 10, the organic light emitting device using the charge generation layer material of the 2-stack white organic light emitting device of the present disclosure had improved light emission efficiency and lifetime compared to Comparative Examples 2-1 to 2-6.

(163) Such a result is considered to be due to the fact that the compound of the present disclosure used as an N-type charge generation layer formed with the disclosed skeleton having proper length and strength, and flatness and a proper hetero-compound capable of binding to metals forms a gap state in the N-type charge generation layer by doping an alkali metal or an alkaline earth metal thereto, and electrons produced from a P-type charge generation layer are readily injected into an electron transfer layer through the gap state produced in the N-type charge generation layer. Accordingly, the P-type charge generation layer may favorably inject and transfer electrons to the N-type charge generation layer, and as a result, driving voltage was lowered, and efficiency and lifetime were improved in the organic light emitting device.

Experimental Example 3

(164) 1) Manufacture of Organic Light Emitting Device

(165) A transparent ITO electrode thin film obtained from glass for an OLED (manufactured by Samsung-Corning Co., Ltd.) was ultrasonic cleaned using trichloroethylene, acetone, ethanol and distilled water consecutively for 5 minutes each, stored in isopropanol, and used.

(166) Next, the ITO substrate was installed in a substrate folder of a vacuum deposition apparatus, and the following 4,4',4''-tris(N,N-(2-naphthyl)-phenylamino)triphenylamine (2-TNATA) was introduced to a cell in the vacuum deposition apparatus.

(167) ##STR00616##

(168) Subsequently, the chamber was evacuated until the degree of vacuum therein reached 10.sup.-6 torr, and then 2-TNATA was evaporated by applying a current to the cell to deposit a hole injection layer having a thickness of 600 Å on the ITO substrate.

(169) To another cell in the vacuum deposition apparatus, the following N,N'-bis(α-naphthyl)-N,N'-diphenyl-4,4'-diamine (NPB) was introduced, and evaporated by applying a current to the cell to deposit a hole transfer layer having a thickness of 300 Å on the hole injection layer.

(170) ##STR00617##

(171) After forming the hole injection layer and the hole transfer layer as above, a blue light emitting material having a structure as below was deposited thereon as a light emitting layer. Specifically, in one side cell in the vacuum deposition apparatus, H1, a blue light emitting host material, was vacuum deposited to a thickness of 200 Å, and D1, a blue light emitting dopant material, was vacuum deposited thereon by 5% with respect to the host material.

(172) ##STR00618##

(173) Subsequently, a compound of the following Table 11 was deposited to a thickness of 300 Å as an electron transfer layer.

(174) ##STR00619##

(175) As an electron injection layer, lithium fluoride (LiF) was deposited to a thickness of 10 Å, and an Al cathode was employed to a thickness of 1,000 Å, and as a result, an OLED was manufactured.

(176) Meanwhile, all the organic compounds required to manufacture the OLED were vacuum sublimation purified under 10.sup.-8 torr to 10.sup.-6 torr by each material to be used in the OLED manufacture.

(177) 2) Driving Voltage and Light Emission Efficiency of Organic Light Emitting Device

(178) For each of the organic light emitting devices manufactured as above, electroluminescent (EL) properties were measured using M7000 manufactured by McScience Inc., and with the measurement results, T.sub.95 was measured when standard luminance was 700 cd/m.sup.2 through a lifetime measurement system (M6000) manufactured by McScience Inc. Results of measuring driving voltage, light emission efficiency, external quantum efficiency and color coordinate (CIE) of the white organic light emitting devices manufactured according to the present disclosure are as shown in Table 11.

(179) TABLE-US-00011 TABLE 11 Light Driving Emission Life- Com- Voltage Efficiency time
pound (V) (cd/A) CIE (x, y) (T.sub.95) Example 3-1 1 5.50 6.32 (0.134, 0.101) 33 Example 3-2 4
5.44 6.44 (0.134, 0.102) 32 Example 3-3 8 5.34 6.38 (0.134, 0.101) 31 Example 3-4 12 5.38 6.20
(0.134, 0.103) 35 Example 3-5 15 5.60 6.12 (0.134, 0.102) 34 Example 3-6 18 5.45 6.01 (0.134,
0.101) 33 Example 3-7 21 5.44 6.22 (0.134, 0.102) 30 Example 3-8 25 5.46 6.18 (0.134, 0.101) 29
Example 3-9 28 5.40 6.13 (0.134, 0.101) 31 Example 3-10 32 5.44 6.30 (0.134, 0.100) 32 Example
3-11 35 5.37 6.35 (0.134, 0.101) 32 Example 3-12 38 5.38 6.41 (0.134, 0.100) 30 Example 3-13 41
4.81 6.93 (0.134, 0.100) 30 Example 3-14 44 5.48 6.21 (0.134, 0.100) 87 Example 3-15 47 5.47
6.28 (0.134, 0.100) 79 Example 3-16 50 4.45 6.98 (0.134, 0.100) 42 Example 3-17 53 4.52 6.63
(0.134, 0.102) 44 Example 3-18 58 5.12 6.20 (0.134, 0.101) 40 Example 3-19 61 4.39 6.87 (0.134,
0.102) 43 Example 3-20 63 4.71 6.48 (0.134, 0.100) 34 Example 3-21 67 4.38 7.00 (0.134, 0.103)
39 Example 3-22 73 5.34 6.31 (0.134, 0.100) 36 Example 3-23 76 5.40 6.36 (0.134, 0.102) 32
Example 3-24 78 5.42 6.26 (0.134, 0.101) 32 Example 3-25 81 5.39 6.19 (0.134, 0.100) 31

Example 3-26 83 5.55 6.27 (0.134, 0.102) 31 Example 3-27 86 4.98 6.20 (0.134, 0.103) 30
Example 3-28 90 5.41 6.19 (0.134, 0.100) 32 Example 3-29 92 5.22 6.28 (0.134, 0.103) 36
Example 3-30 95 5.19 6.41 (0.134, 0.102) 33 Example 3-31 101 5.43 6.17 (0.134, 0.100) 62
Example 3-32 103 4.94 6.85 (0.134, 0.099) 40 Example 3-33 107 4.51 6.97 (0.134, 0.102) 42
Example 3-34 110 4.45 7.03 (0.134, 0.100) 39 Example 3-35 113 5.00 6.21 (0.134, 0.103) 32
Example 3-36 115 4.93 6.35 (0.134, 0.101) 35 Example 3-37 122 4.50 6.92 (0.134, 0.104) 41
Example 3-38 124 5.03 6.27 (0.134, 0.100) 35 Example 3-39 126 5.11 6.22 (0.134, 0.103) 36
Example 3-40 130 5.01 6.98 (0.134, 0.100) 31 Example 3-41 135 4.98 7.10 (0.134, 0.103) 33
Example 3-42 138 5.34 6.21 (0.134, 0.100) 36 Example 3-43 142 5.40 6.36 (0.134, 0.102) 39
Example 3-44 144 5.42 6.26 (0.134, 0.101) 32 Example 3-45 147 5.39 6.19 (0.134, 0.100) 31
Example 3-46 149 5.55 6.27 (0.134, 0.102) 31 Example 3-47 152 4.98 6.20 (0.134, 0.103) 30
Example 3-48 155 5.41 6.19 (0.134, 0.100) 32 Example 3-49 161 5.22 6.28 (0.134, 0.103) 36
Example 3-50 164 5.19 6.41 (0.134, 0.102) 33 Example 3-51 168 5.43 6.17 (0.134, 0.100) 62
Example 3-52 172 4.94 6.85 (0.134, 0.099) 40 Example 3-53 174 4.51 6.97 (0.134, 0.102) 42
Example 3-54 178 4.45 7.03 (0.134, 0.100) 39 Example 3-55 181 5.00 6.21 (0.134, 0.103) 32
Example 3-56 186 5.43 6.17 (0.134, 0.100) 62 Example 3-57 190 4.94 6.85 (0.134, 0.099) 40
Example 3-58 194 4.51 6.97 (0.134, 0.102) 42 Example 3-59 198 4.45 7.03 (0.134, 0.100) 39
Example 3-60 201 5.00 6.21 (0.134, 0.103) 32 Example 3-61 202 4.93 6.35 (0.134, 0.101) 35
Example 3-62 206 4.50 6.92 (0.134, 0.104) 41 Example 3-63 210 5.03 6.27 (0.134, 0.100) 35
Example 3-64 214 5.11 6.22 (0.134, 0.103) 36 Example 3-65 218 5.01 6.98 (0.134, 0.100) 31
Example 3-66 221 4.98 7.10 (0.134, 0.103) 33 Example 3-67 222 5.34 6.21 (0.134, 0.100) 36
Example 3-68 226 5.40 6.36 (0.134, 0.102) 39 Example 3-69 230 5.42 6.26 (0.134, 0.101) 32
Example 3-70 234 5.22 6.28 (0.134, 0.103) 36 Example 3-71 238 5.19 6.41 (0.134, 0.102) 33
Example 3-72 241 5.43 6.17 (0.134, 0.100) 62 Example 3-73 242 4.94 6.85 (0.134, 0.099) 40
Example 3-74 246 4.51 6.97 (0.134, 0.102) 42 Example 3-75 250 4.45 7.03 (0.134, 0.100) 39
Example 3-76 254 5.00 6.21 (0.134, 0.103) 32 Example 3-77 258 5.43 6.17 (0.134, 0.100) 62
Example 3-78 261 4.94 6.85 (0.134, 0.099) 40 Example 3-79 262 4.51 6.97 (0.134, 0.102) 42
Example 3-80 266 4.45 7.03 (0.134, 0.100) 39 Example 3-81 270 5.00 6.21 (0.134, 0.103) 32
Example 3-82 274 4.93 6.35 (0.134, 0.101) 35 Example 3-83 278 4.94 6.85 (0.134, 0.099) 40
Example 3-84 280 4.51 6.97 (0.134, 0.102) 42 Example 3-85 282 4.45 7.03 (0.134, 0.100) 39
Example 3-86 286 5.00 6.21 (0.134, 0.103) 32 Example 3-87 290 4.93 6.35 (0.134, 0.101) 35
Example 3-88 294 4.50 6.92 (0.134, 0.104) 41 Example 3-89 298 5.03 6.27 (0.134, 0.100) 35
Example 3-90 301 5.11 6.22 (0.134, 0.103) 36 Example 3-91 302 5.01 6.98 (0.134, 0.100) 31
Example 3-92 306 4.98 7.10 (0.134, 0.103) 33 Example 3-93 310 5.01 6.98 (0.134, 0.100) 31
Example 3-94 314 4.98 7.10 (0.134, 0.103) 33 Example 3-95 318 5.34 6.21 (0.134, 0.100) 36
Example 3-96 321 5.40 6.36 (0.134, 0.102) 39 Example 3-97 322 5.42 6.26 (0.134, 0.101) 32
Example 3-98 326 5.22 6.28 (0.134, 0.103) 36 Example 3-99 330 5.19 6.41 (0.134, 0.102) 33
Example 3-100 334 5.43 6.17 (0.134, 0.100) 62 Example 3-101 338 4.94 6.85 (0.134, 0.099) 40
Example 3-102 341 4.51 6.97 (0.134, 0.102) 42 Example 3-103 342 4.45 7.03 (0.134, 0.100) 39
Example 3-104 346 5.00 6.21 (0.134, 0.103) 32 Example 3-105 350 5.43 6.17 (0.134, 0.100) 62
Example 3-106 354 5.22 6.28 (0.134, 0.103) 36 Example 3-107 358 5.19 6.41 (0.134, 0.102) 33
Example 3-108 361 5.43 6.17 (0.134, 0.100) 62 Example 3-109 362 4.94 6.85 (0.134, 0.099) 40
Example 3-110 366 4.51 6.97 (0.134, 0.102) 42 Example 3-111 370 4.45 7.03 (0.134, 0.100) 39
Example 3-112 374 5.00 6.21 (0.134, 0.103) 32 Example 3-113 378 5.43 6.17 (0.134, 0.100) 62
Example 3-114 381 4.94 6.85 (0.134, 0.099) 40 Example 3-115 382 4.51 6.97 (0.134, 0.102) 42
Example 3-116 386 4.45 7.03 (0.134, 0.100) 39 Example 3-117 390 5.00 6.21 (0.134, 0.103) 32
Example 3-118 394 4.93 6.35 (0.134, 0.101) 35 Example 3-119 398 4.50 6.92 (0.134, 0.104) 41
Example 3-120 401 5.03 6.27 (0.134, 0.100) 35 Example 3-121 402 5.48 6.21 (0.134, 0.100) 87
Example 3-122 406 5.47 6.28 (0.134, 0.100) 79 Example 3-123 410 4.45 6.98 (0.134, 0.100) 42
Example 3-124 414 4.52 6.63 (0.134, 0.102) 44 Example 3-125 418 5.12 6.20 (0.134, 0.101) 40

Example 3-126 421 4.39 6.87 (0.134, 0.102) 43 Example 3-127 422 4.71 6.48 (0.134, 0.100) 34
Example 3-128 426 4.38 7.00 (0.134, 0.103) 39 Example 3-129 430 5.34 6.31 (0.134, 0.100) 36
Example 3-130 446 5.40 6.36 (0.134, 0.102) 32 Example 3-131 450 5.42 6.26 (0.134, 0.101) 32
Example 3-132 454 5.39 6.19 (0.134, 0.100) 31 Example 3-133 458 5.55 6.27 (0.134, 0.102) 31
Example 3-134 462 4.98 6.20 (0.134, 0.103) 30 Example 3-135 466 5.41 6.19 (0.134, 0.100) 32
Example 3-136 470 5.22 6.28 (0.134, 0.103) 36 Example 3-137 474 5.19 6.41 (0.134, 0.102) 33
Example 3-138 478 5.43 6.17 (0.134, 0.100) 62 Example 3-139 482 4.94 6.85 (0.134, 0.099) 40
Example 3-140 486 4.51 6.97 (0.134, 0.102) 42 Example 3-141 490 4.45 7.03 (0.134, 0.100) 39
Example 3-142 494 5.00 6.21 (0.134, 0.103) 32 Example 3-143 498 4.93 6.35 (0.134, 0.101) 35
Example 3-144 502 4.50 6.92 (0.134, 0.104) 41 Example 3-145 506 5.03 6.27 (0.134, 0.100) 35
Example 3-146 510 5.11 6.22 (0.134, 0.103) 36 Example 3-147 514 5.01 6.98 (0.134, 0.100) 31
Example 3-148 518 4.98 7.10 (0.134, 0.103) 33 Example 3-149 522 5.34 6.21 (0.134, 0.100) 36
Example 3-150 526 5.40 6.36 (0.134, 0.102) 39 Example 3-151 530 5.42 6.26 (0.134, 0.101) 32
Example 3-152 534 5.39 6.19 (0.134, 0.100) 31 Example 3-153 538 5.55 6.27 (0.134, 0.102) 31
Example 3-154 542 4.98 6.20 (0.134, 0.103) 30 Example 3-155 546 5.41 6.19 (0.134, 0.100) 32
Example 3-156 550 5.22 6.28 (0.134, 0.103) 36 Example 3-157 554 5.19 6.41 (0.134, 0.102) 33
Example 3-158 558 5.43 6.17 (0.134, 0.100) 62 Example 3-159 562 4.94 6.85 (0.134, 0.099) 40
Example 3-160 566 4.51 6.97 (0.134, 0.102) 42 Example 3-161 570 4.45 7.03 (0.134, 0.100) 39
Example 3-162 574 5.00 6.21 (0.134, 0.103) 32 Example 3-163 578 5.43 6.17 (0.134, 0.100) 62
Example 3-164 582 4.94 6.85 (0.134, 0.099) 40 Example 3-165 586 5.43 6.17 (0.134, 0.100) 62
Example 3-166 590 4.94 6.85 (0.134, 0.099) 40 Example 3-167 594 4.51 6.97 (0.134, 0.102) 42
Example 3-168 598 4.45 7.03 (0.134, 0.100) 39 Example 3-169 602 5.00 6.21 (0.134, 0.103) 32
Example 3-170 606 5.43 6.17 (0.134, 0.100) 62 Example 3-171 610 4.94 6.85 (0.134, 0.099) 40
Example 3-172 614 4.51 6.97 (0.134, 0.102) 42 Example 3-173 618 4.45 7.03 (0.134, 0.100) 39
Example 3-174 622 5.00 6.21 (0.134, 0.103) 32 Example 3-175 626 4.93 6.35 (0.134, 0.101) 35
Example 3-176 630 4.50 6.92 (0.134, 0.104) 41 Example 3-177 634 5.03 6.27 (0.134, 0.100) 35
Example 3-178 638 5.48 6.21 (0.134, 0.100) 87 Example 3-179 642 5.47 6.28 (0.134, 0.100) 79
Example 3-180 646 4.45 6.98 (0.134, 0.100) 42 Example 3-181 650 4.52 6.63 (0.134, 0.102) 44
Example 3-182 654 5.12 6.20 (0.134, 0.101) 40 Example 3-183 658 4.39 6.87 (0.134, 0.102) 43
Example 3-184 662 4.71 6.48 (0.134, 0.100) 34 Example 3-185 666 5.22 6.28 (0.134, 0.103) 36
Example 3-186 670 5.19 6.41 (0.134, 0.102) 33 Example 3-187 674 5.43 6.17 (0.134, 0.100) 62
Example 3-188 678 4.94 6.85 (0.134, 0.099) 40 Example 3-189 682 4.51 6.97 (0.134, 0.102) 42
Example 3-190 686 4.45 7.03 (0.134, 0.100) 39 Example 3-191 690 5.00 6.21 (0.134, 0.103) 32
Example 3-192 694 5.43 6.17 (0.134, 0.100) 62 Example 3-193 698 4.94 6.85 (0.134, 0.099) 40
Example 3-194 702 5.43 6.17 (0.134, 0.100) 62 Example 3-195 706 4.94 6.85 (0.134, 0.099) 40
Example 3-196 710 4.51 6.97 (0.134, 0.102) 42 Example 3-197 714 5.01 6.98 (0.134, 0.100) 31
Example 3-198 718 4.98 7.10 (0.134, 0.103) 33 Example 3-199 722 5.34 6.21 (0.134, 0.100) 36
Example 3-200 726 5.40 6.36 (0.134, 0.102) 39 Example 3-201 730 5.42 6.26 (0.134, 0.101) 32
Example 3-202 734 5.39 6.19 (0.134, 0.100) 31 Example 3-203 738 5.55 6.27 (0.134, 0.102) 31
Example 3-204 742 4.98 6.20 (0.134, 0.103) 30 Example 3-205 746 5.41 6.19 (0.134, 0.100) 32
Example 3-206 750 5.22 6.28 (0.134, 0.103) 36 Example 3-207 754 4.51 6.97 (0.134, 0.102) 42
Example 3-208 758 4.45 7.03 (0.134, 0.100) 39 Example 3-209 762 5.00 6.21 (0.134, 0.103) 32
Example 3-210 766 4.93 6.35 (0.134, 0.101) 35 Example 3-211 770 4.50 6.92 (0.134, 0.104) 41
Example 3-212 774 5.03 6.27 (0.134, 0.100) 35 Example 3-213 778 5.11 6.22 (0.134, 0.103) 36
Example 3-214 782 5.01 6.98 (0.134, 0.100) 31 Example 3-215 786 4.98 7.10 (0.134, 0.103) 33
Example 3-216 790 5.01 6.98 (0.134, 0.100) 31 Example 3-217 794 4.98 7.10 (0.134, 0.103) 33
Example 3-218 898 5.34 6.21 (0.134, 0.100) 36 Example 3-219 802 5.22 6.28 (0.134, 0.103) 36
Example 3-220 806 5.19 6.41 (0.134, 0.102) 33 Example 3-221 810 5.43 6.17 (0.134, 0.100) 62
Example 3-222 814 4.94 6.85 (0.134, 0.099) 40 Example 3-223 818 4.51 6.97 (0.134, 0.102) 42
Example 3-224 822 4.50 6.92 (0.134, 0.104) 41 Example 3-225 826 5.43 6.17 (0.134, 0.100) 62

Example 3-226 830 4.94 6.85 (0.134, 0.099) 40 Comparative A 2.82 2.23 (0.134, 0.110) 15
Example 3-1 Comparative B 3.80 4.32 (0.134, 0.111) 20 Example 3-2 Comparative C 1.78 3.39
(0.128, 0.099) 11 Example 3-3 Comparative D 3.56 1.23 (0.119, 0.100) 9 Example 3-4
Comparative E 1.89 2.23 (0.128, 0.111) 10 Example 3-5 Comparative F 2.13 1.30 (0.125, 0.097) 11
Example 3-6

(180) As seen from the results of Table 11, the organic light emitting device using the electron transfer layer material of the white organic light emitting device of the present disclosure had significantly improved light emission efficiency and lifetime compared to Comparative Examples 3-1 to 3-6.

(181) Such a result is considered to be due to the fact that, when using the disclosed compound having proper length and strength, and flatness as an electron transfer layer, a compound in an excited state is made by receiving electrons under a specific condition, and particularly when an excited state is formed in the hetero-skeleton site of the compound, excited energy moves to a stable state before the excited hetero-skeleton site goes through other reactions, and as a result, the relatively stabilized compound is capable of efficiently transferring electrons without the compound being decomposed or destroyed. For reference, those that are stable when excited are considered to be aryl or acene-based compounds or polycyclic hetero-compounds.

(182) Accordingly, it is considered that the compound of the present disclosure brought excellence in all aspects of driving, efficiency and lifetime by enhancing improved electron-transfer properties or improved stability.

Claims

1. A heterocyclic compound represented by the following Chemical Formula 1: ##STR00620## wherein, in Chemical Formula 1, X is O or S, L is a direct bond; a substituted or unsubstituted C6 to C30 arylene group; or a substituted or unsubstituted C2 to C30 heteroarylene group, Z is a substituted or unsubstituted C6 to C30 aryl group; a substituted or unsubstituted C2 to C30 heteroaryl group or a substituted or unsubstituted phosphine oxide group, R1 is a substituted or unsubstituted C6 to C30 aryl group; or a substituted or unsubstituted C2 to C30 heteroaryl group, R2 to R5 are each independently hydrogen or deuterium, a is an integer of 1 to 4, b is an integer of 1 to 3, m and n are each an integer of 1 to 5, and when a, b, m and n are each 2 or greater, substituents in the parentheses are the same as or different from each other.
2. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by the following Chemical Formula 2 or 3: ##STR00621## in Chemical Formulae 2 and 3, L, Z, R1, R4, R5, a, b, m and n have the same definitions as in Chemical Formula 1.
3. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by any one of the following Chemical Formulae 4 to 6: ##STR00622## in Chemical Formulae 4 to 6, X, L, Z, R1 to R5, a, b, m and n have the same definitions as in Chemical Formula 1.
4. The heterocyclic compound of claim 1, wherein R1 is a substituted or unsubstituted C6 to C30 aryl group.
5. The heterocyclic compound of claim 1, wherein L is a direct bond; a phenylene group; a biphenylene group; a terphenylene group; a naphthylene group; an anthracenylene group; a triphenylenylene group; or a phenanthrenylene group.
6. The heterocyclic compound of claim 1, wherein Z is a substituted or unsubstituted phenyl group; a biphenyl group; a terphenyl group; a naphthyl group; a triphenylenyl group; a phenanthrenyl group; an anthracenyl group; a dimethylfluorenyl group; a diphenylfluorenyl group; a spirobifluorenyl group; an isoquinolinyl group; a quinazolinyl group; a phenoxazinyl group; a phenothiazinyl group; an indolocarbazole group; a benzonaphthothiophene group; a substituted or unsubstituted pyrazole group; a substituted or unsubstituted pyridine group; a substituted or unsubstituted pyrimidine group; a substituted or unsubstituted triazine group; a substituted or

unsubstituted carbazole group; a substituted or unsubstituted phenanthrolinyl group; a substituted or unsubstituted dibenzofuran group; a substituted or unsubstituted dibenzothiophene group; a substituted or unsubstituted dihydroacridine group; or a substituted or unsubstituted phosphine oxide group.

7. The heterocyclic compound of claim 1, wherein Chemical Formula 1 is represented by any one of the following compounds: ##STR00623## ##STR00624## ##STR00625## ##STR00626## ##STR00627## ##STR00628## ##STR00629## ##STR00630## ##STR00631## ##STR00632## ##STR00633## ##STR00634## ##STR00635## ##STR00636## ##STR00637## ##STR00638## ##STR00639## ##STR00640## ##STR00641## ##STR00642## ##STR00643## ##STR00644## ##STR00645## ##STR00646## ##STR00647## ##STR00648## ##STR00649## ##STR00650## ##STR00651## ##STR00652## ##STR00653## ##STR00654## ##STR00655## ##STR00656## ##STR00657## ##STR00658## ##STR00659## ##STR00660## ##STR00661## ##STR00662## ##STR00663## ##STR00664## ##STR00665## ##STR00666## ##STR00667## ##STR00668## ##STR00669## ##STR00670## ##STR00671## ##STR00672## ##STR00673## ##STR00674## ##STR00675## ##STR00676## ##STR00677## ##STR00678## ##STR00679## ##STR00680## ##STR00681## ##STR00682## ##STR00683## ##STR00684## ##STR00685## ##STR00686## ##STR00687## ##STR00688## ##STR00689## ##STR00690## ##STR00691## ##STR00692## ##STR00693## ##STR00694## ##STR00695## ##STR00696## ##STR00697## ##STR00698## ##STR00699## ##STR00700## ##STR00701## ##STR00702## ##STR00703## ##STR00704## ##STR00705## ##STR00706## ##STR00707## ##STR00708## ##STR00709## ##STR00710## ##STR00711## ##STR00712## ##STR00713## ##STR00714## ##STR00715## ##STR00716## ##STR00717## ##STR00718## ##STR00719## ##STR00720## ##STR00721## ##STR00722## ##STR00723## ##STR00724## ##STR00725## ##STR00726## ##STR00727## ##STR00728## ##STR00729## ##STR00730## ##STR00731## ##STR00732## ##STR00733## ##STR00734## ##STR00735## ##STR00736## ##STR00737## ##STR00738## ##STR00739## ##STR00740## ##STR00741## ##STR00742## ##STR00743## ##STR00744## ##STR00745## ##STR00746## ##STR00747## ##STR00748## ##STR00749## ##STR00750## ##STR00751## ##STR00752## ##STR00753## ##STR00754## ##STR00755## ##STR00756## ##STR00757## ##STR00758## ##STR00759## ##STR00760## ##STR00761## ##STR00762## ##STR00763## ##STR00764## ##STR00765## ##STR00766## ##STR00767## ##STR00768## ##STR00769## ##STR00770## ##STR00771## ##STR00772## ##STR00773## ##STR00774## ##STR00775## ##STR00776## ##STR00777## ##STR00778## ##STR00779## ##STR00780## ##STR00781## ##STR00782## ##STR00783## ##STR00784## ##STR00785## ##STR00786## ##STR00787## ##STR00788## ##STR00789## ##STR00790## ##STR00791## ##STR00792## ##STR00793## ##STR00794## ##STR00795## ##STR00796## ##STR00797## ##STR00798## ##STR00799## ##STR00800## ##STR00801## ##STR00802##

8. An organic light emitting device comprising: a first electrode; a second electrode; and an organic material layer provided between the first electrode and the second electrode, wherein the organic material layer includes the heterocyclic compound of claim 1.

9. The organic light emitting device of claim 8, wherein the organic material layer includes an electron transfer layer, and the electron transfer layer includes the heterocyclic compound.

10. The organic light emitting device of claim 8, wherein the organic material layer includes a charge generation layer, and the charge generation layer includes the heterocyclic compound.

11. The organic light emitting device of claim 8, further comprising one, two or more layers selected from the group consisting of a light emitting layer, a hole injection layer, a hole transfer layer, an electron injection layer, an electron transfer layer, an electron blocking layer and a hole blocking layer.

12. The organic light emitting device of claim 8 comprising: the first electrode; a first stack provided on the first electrode and including a first light emitting layer; a charge generation layer provided on the first stack; a second stack provided on the charge generation layer and including a

second light emitting layer; and the second electrode provided on the second stack.

13. The organic light emitting device of claim 12, wherein the charge generation layer includes the heterocyclic compound.

14. The organic light emitting device of claim 12, wherein the charge generation layer includes an N-type charge generation layer, and the N-type charge generation layer includes the heterocyclic compound.
